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Décomposition de HODGE–DE RHAM sur une surface fermée de $\mathbb{R}^3$: discrétisation DDFV

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Résumé

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Mots-clés.

Abstract

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Keywords.

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Intro : seuls deux papiers généralisent en 3D, le formalisme et la technique de preuve sont nouveaux.

Part I

Convergence analysis of the DDFV method on surfaces in $\mathbb{R}^3$

Chapter I

Generalization of the DDFV method to 2D surfaces in $\mathbb{R}^3$

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[TMR16; TM19] [ABH07, Sec. 2.1]

I.1 Hypothesis on the surface & Projection

I.1.1 Global coordinates

[DE13, Sec. 2.3] It is quite convenient to use global coordinates in a neighbourhood of a hypersurface (see [DE13, Sec. 2.2] for a definition of *hypersurfaces*), the so-called *Fermi coordinates*. This avoids working with charts and atlases when proving results and carrying out the numerical analysis. For this one introduces the *oriented distance function* for $\mathcal{S}$.

Assume in the following that $\mathcal{O} \subset \mathbb{R}^3$ is bounded and open with exterior normal $\mathbf{n}$ and assume that $\mathcal{S} = \partial\mathcal{O}$ is a $\mathcal{C}^k$ -hypersurface ($k \geq 2$). The *oriented distance function* for $\mathcal{S}$ is defined by

$$d(x) \stackrel{\text{d\'ef.}}{=} \begin{cases} \inf_{y \in \mathcal{S}} |x - y| & x \in \mathbb{R}^3 \setminus \overline{\mathcal{O}} \\ -\inf_{y \in \mathcal{S}} |x - y| & x \in \mathcal{O} \end{cases}.$$

One easily verifies that d is globally Lipschitz-continuous with Lipschitz constant 1. Since $\partial\mathcal{O}$ is a $\mathcal{C}^2$ -hypersurface, it satisfies both a uniform interior and a uniform exterior sphere condition, which means that for each point $x_0 \in \partial\Omega$ there are balls B and B' such that

$$\overline{B} \cap (\mathbb{R}^3 \setminus \Omega) = \{x_0\}, \quad \overline{B'} \cap \overline{\Omega} = \{x_0\},$$

and the radii of B , B' are bounded from below by a positive constant δ uniformly in x_0 . With this observation the following lemma is **easily proved**.

Lemma I.1.1 ([DE13, Lemma 2.8]). *We define*

$$U_\delta \stackrel{\text{d\'ef.}}{=} \{x \in \mathbb{R}^3 \mid |d(x)| < \delta\}.$$

Then $d \in \mathcal{C}^k(U_\delta)$, and for every point $x \in U_\delta$ there exists a unique point $\Pi^\mathcal{S}(x) \in \mathcal{S}$ such that

$$x = \Pi^\mathcal{S}(x) + d(x) \mathbf{n}(\Pi^\mathcal{S}(x)). \quad (\text{I.1.1})$$

Moreover, we have that

$$\nabla d(x) = \mathbf{n}(\Pi^\mathcal{S}(x)), \quad |\nabla d(x)| = 1, \quad \forall x \in U_\delta.$$

We also extend the normal constantly in the normal direction: $\mathbf{n}(x) = \mathbf{n}(\Pi^\mathcal{S}(x))$ for $x \in U_\delta$. Thus we have introduced a global coordinate system around $\mathcal{S}$. Every point $x \in U_\delta$ can be described by its Fermi coordinates (normal coordinates) $d(x)$ and $\Pi^\mathcal{S}(x)$ according to Equation (I.1.1).

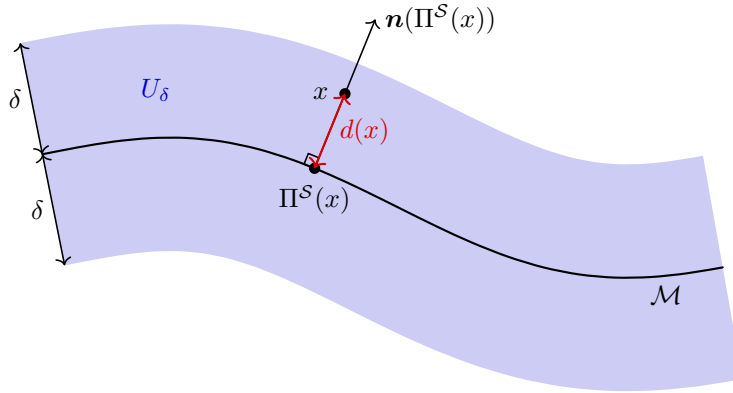


Figure I.1. Strip U_δ around the hypersurface $\mathcal{S}$ and normal coordinates $x = \Pi^\mathcal{S}(x) + d(x) \nu(\Pi^\mathcal{S}(x))$

Proposition I.1.2. *The map $\Pi^\mathcal{S} : \mathcal{S} \rightarrow \mathcal{S}$ is bijective.*

I.1.2 Sobolev spaces on surfaces

[DE13, p. 299] Let $\mathcal{S} \in \mathcal{C}^2$ for the following. For $p \in [1; \infty]$ we let $L^p(\mathcal{S})$ denote the space of functions $f : \mathcal{S} \rightarrow \mathbb{R}$ which are measurable with respect to the surface measure dA (dA in connection with an integral over $\mathcal{S}$ denotes the n -dimensional Hausdorff measure) and have finite norm, where the norm is defined by

$$\|f\|_{L^p(\mathcal{S})} \stackrel{\text{d\'ef.}}{=} \left(\int_{\mathcal{S}} |f|^p dA \right)^{1/p}$$

for $p < \infty$, and for $p = \infty$ we mean the essential supremum norm. The space $L^p(\mathcal{S})$ is a Banach space and for $p = 2$ a Hilbert space. For $1 \leq p < \infty$ the space $\mathcal{C}^0(\mathcal{S})$ and $\mathcal{C}^1(\mathcal{S})$ are dense in $L^p(\mathcal{S})$.

Theorem I.1.3 (Poincaré's inequality [DE13, Sec. 2.4]). Assume that $\mathcal{S} \in \mathcal{C}^3$ and $1 \leq p < \infty$. Then there is a constant c such that, for every function $f \in H^{1,p}(\mathcal{S})$ with $\int_{\mathcal{S}} f \, dA = 0$, we have the inequality

$$\|f\|_{L^p(\mathcal{S})} \leq c \|\nabla_{\mathcal{S}} f\|_{L^p(\mathcal{S})}.$$

I.1.3 Triangulations [DE13, Sec. 4.1]

The smooth 2-dimensional surface $\mathcal{S}$ ($\partial\mathcal{S} = \emptyset$) is approximated by a surface S which lies in the strip U_{δ} (see Lemma I.1.1) and which is a Lipschitz surface. The strip can be chosen with locally varying width: see Equation (I.1.3).

We assume that the discrete triangulated surface S besides Equation (I.1.3) has the following properties. The surface S is the union of finitely many non-degenerate (closed) 2-simplices in $\mathbb{R}^3$. We let $\mathbf{M}$ denote the set of these simplices:

$$S = \bigcup_{K \in \mathbf{M}} K.$$

The vertices $\mathfrak{B}^*$ of the simplices are taken to sit on the smooth surface $\mathcal{S}$. For $K, L \in \mathbf{M}$, either $K \cap L = \emptyset$ or $K \cap L$ is an $(3 - k)$ -dimensional side simplex ($k \in \{1, 2, 3\}$) common to both of the simplices K and L . For $K \in \mathbf{M}$ we denote by d_K its diameter and by $r(K)$ the in-ball radius. Also, we define

$$h_{\mathbf{M}} \stackrel{\text{def.}}{=} \max_{K \in \mathbf{M}} d_K \quad \text{and} \quad r_{\mathbf{M}} \stackrel{\text{def.}}{=} \min_{K \in \mathbf{M}} r(K). \quad (\text{I.1.2})$$

We assume that the quantity

$$\sigma_{\mathbf{M}} \stackrel{\text{def.}}{=} \max_{K \in \mathbf{M}} \sigma(K) \quad \text{with} \quad \sigma(K) = \frac{h(K)}{r(K)}$$

is uniformly bounded independently of $h_{\mathbf{M}}$.

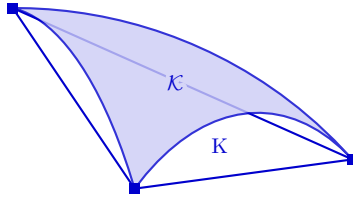


Figure I.2. Approximation of a smooth surface by a polygonal surface : a control volume K and its projection $\kappa = \Pi^{\mathcal{S}} K$ onto the hyperurface $\mathcal{S}$

Note that, by Lemma I.1.1, for every simplex $K \subset S$ there is a unique curved simplex $\kappa \stackrel{\text{def.}}{=} \Pi^{\mathcal{S}} K \subset \mathcal{S}$ (see Figure I.2). In order to avoid global double covering, we assume that for each point $a \in \mathcal{S}$ there is at most one point $x \in S$ with $a = \Pi^{\mathcal{S}}(x)$. This implies that there is a bijective correspondence between the triangles on S and the induced curvilinear triangles on $\mathcal{S}$.

We assume that $\mathcal{S}$ has only one connected component. We start with a discrete surface S contained in the variable strip U_{δ} ,

$$S \subset U_{\delta} = \{y + s\mathbf{n}(y) \mid |s| < \delta(y), y \in \mathcal{S}\}, \quad (\text{I.1.3})$$

where for a point $y \in \mathcal{S}$ we set

$$\delta(y) = \min \left\{ \left(\max_{i=1, \dots, n} |\kappa_i(y)| \right)^{-1}, \sup_{z \in \mathcal{S}, z \neq y} \frac{L(y, z)}{|y - z|} \right\}$$

with the sectional curvatures κ_i and where $L(y, z)$ denotes the geodesic distance between y and z on $\mathcal{S}$ (see [Lee97] for the definitions of *sectional curvature* and *geodesic distance*).

Remark I.1.4. We can take $\delta \stackrel{\text{def.}}{=} \min_{y \in \mathcal{S}} \delta(y)$ which exists since $\mathcal{S}$ is a compact manifold.

Proposition I.1.5. La courbure de la surface étant finie en tout point, il existe toujours δ pour que U_{δ} ne s'auto-intersecte pas. Nous faisons l'hypothèse

$\mathcal{H}$ Il existe h suffisamment petit tel que $S \subset U_\delta$.

Proposition I.1.6. One can show that each point $x \in U_\delta$ can be uniquely projected onto the smooth surface, yielding the decomposition

$$x = \Pi^S(x) + d(x) \mathbf{n}(\Pi^S(x)), \quad x \in U_\delta,$$

with $\Pi^S(x)$ the orthogonal projection of x onto S and $d(x)$ the oriented distance between x and S .

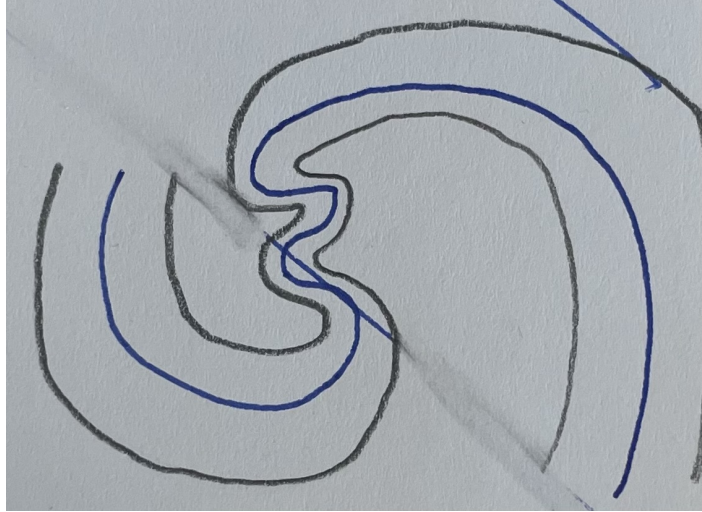


Figure I.3. $\delta(y)$

Theorem I.1.7.

$$\begin{aligned} \kappa &\stackrel{\text{d\'ef.}}{=} \Pi^S K & \kappa^* &\stackrel{\text{d\'ef.}}{=} \Pi^S K^* & \mathcal{D} &\stackrel{\text{d\'ef.}}{=} \Pi^S \mathcal{D} \\ \mathfrak{M} &\stackrel{\text{d\'ef.}}{=} \{\kappa\} & \mathfrak{M}^* &\stackrel{\text{d\'ef.}}{=} \{\kappa^*\} & \mathfrak{D} &\stackrel{\text{d\'ef.}}{=} \{\mathcal{D}\} \end{aligned}$$

I.2 Construction of the DDFV meshes

[ABH07] We consider the class of finite volume schemes introduced by Hermeline in [Her00], by Domelevo, Omnès in [DO05] for the Laplace equation and in [DDO05] for other linear equations like the Stokes or the Div-Curl system. In [DDO05], these schemes are called «Discrete Duality Finite Volume» (DDFV for short) since the discrete gradient and discrete divergence operators are dual one from each other (see I.5.1). The equation is approximated simultaneously on two interrelated meshes: the primal and the dual mesh. The number of variables and of equations doubles compared to usual cell-centered FV schemes, but the gradient approximation (the one already used by Coudière and *al.* [CVV99]) becomes simple and quite efficient.

Construction maillages.

1. Soit une surface S
2. On place des points sur la surface S
3. On en déduit un complexe sur la variété
4. Ce complexe définit le maillage primal, courbe et polyédrique
5. On calcule les barycentres des mailles primales plates
6. On projette ces barycentres sur la surface S
7. Construction des mailles duales (barycentriques):
8. Projection du milieu des arêtes primales sur S

9. Arcs passant par les milieux projetés
10. Les mailles duales sont approchées par $2m$ facettes

I.2.1 Primal mesh

Section I.1 :

$$S = \bigcup_{K \in \mathbf{M}} K, \quad \forall K \subset S, \quad \exists ! K \stackrel{\text{d\'ef.}}{=} \Pi^S K \subset S.$$

The *primal mesh* $\mathfrak{M}$ is the set of disjoint polygonal control volumes $\mathcal{K}$.

$$S = \bigcup_{\mathcal{K} \in \mathfrak{M}} \mathcal{K}.$$

We create a representative point x_K of each triangle $K \in \mathbf{M}$ (I.4, bottom left). The point x_K is taken at the barycenter of triangle K . We denote $x_K = \Pi^S x_K$ its projection onto the smooth surface S .

We call $\mathbf{B}$ this family of points $\{x_K \mid K \in \mathbf{M}\}$ and let $\mathfrak{B}$ denote the set of vertices of the mesh $\mathfrak{M}$. For all adjacent control volumes K and L , we assume that $\partial K \cap \partial L$ is a segment that we call *an edge of the mesh* $\mathbf{M}$ and that we denote by $\varsigma = K \mid L$. Let $\mathbf{E}$ be the set of such edges.

$$\mathfrak{B}^* = \{x_K \mid \mathcal{K} \in \mathfrak{M}\} \quad \forall \varsigma \in \mathbf{E}, \quad \sigma = \Pi^S \varsigma$$

For any control volume $\mathcal{K} \in \mathfrak{M}$, we define

- $m(\mathcal{K})$, the measure of $\mathcal{K}$;
- $\mathfrak{E}_{\mathcal{K}}$, the set of edges of $\mathcal{K}$;
- $\mathfrak{D}_{\mathcal{K}} = \{\mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D} \mid \sigma \in \mathfrak{E}_{\mathcal{K}}\}$;
- $\nu_{\mathcal{K}}$, the outward unit normal vector to $\partial \mathcal{K}$;
- $d_{\mathcal{K}}$, the diameter of $\mathcal{K}$.

It can happen that the point x_K does not belong to $\mathcal{K}$ (if $\mathcal{K}$ is a triangle and x_K its circumcenter for instance). As usual in that case (see [EGH00]) it is necessary to assume that ν_K points from x_K towards x_L , that is $\nu_K = \nu$ with the notations above. Similarly we assume that $\nu_{K^*} = \nu^*$. In the case of a triangular mesh $\mathfrak{M}$ in which x_K is the circumcenter of $\mathcal{K}$, this assumption is known as the Delaunay condition.

I.2.2 Dual mesh (Maillage dual barycentrique)

For all adjacent control volumes $\mathcal{K}$ and $\mathcal{L}$, we denote by $\sigma^* = [x_K, x_L]$ the segment joining their representative points. Let $\mathfrak{E}^*$ be the set of such segments.



Figure I.5

The vertices x_{K^*} and x_K will be referred to as $\mathfrak{M}$ -vertices and $\mathfrak{M}^*$ -vertices respectively. The set $\mathfrak{M}^*$ is a family of «dual» control volumes defined as follows. To any point $x_{K^*} \in \mathfrak{B}$, we associate the polygon $\mathcal{K}^* \in \mathfrak{M}^*$ whose vertices are $\{x_K \in \mathfrak{B}^* \mid x_{K^*} \in \bar{\mathcal{K}}, \mathcal{K} \in \mathfrak{M}\}$ sorted with respect to the clockwise order of the corresponding primal control volumes.

The corresponding notations $\sigma^* = \mathcal{K}^* \mid \mathcal{L}^*$ and $\mathfrak{E}^*$ refer to the dual mesh $\mathfrak{M}^*$.

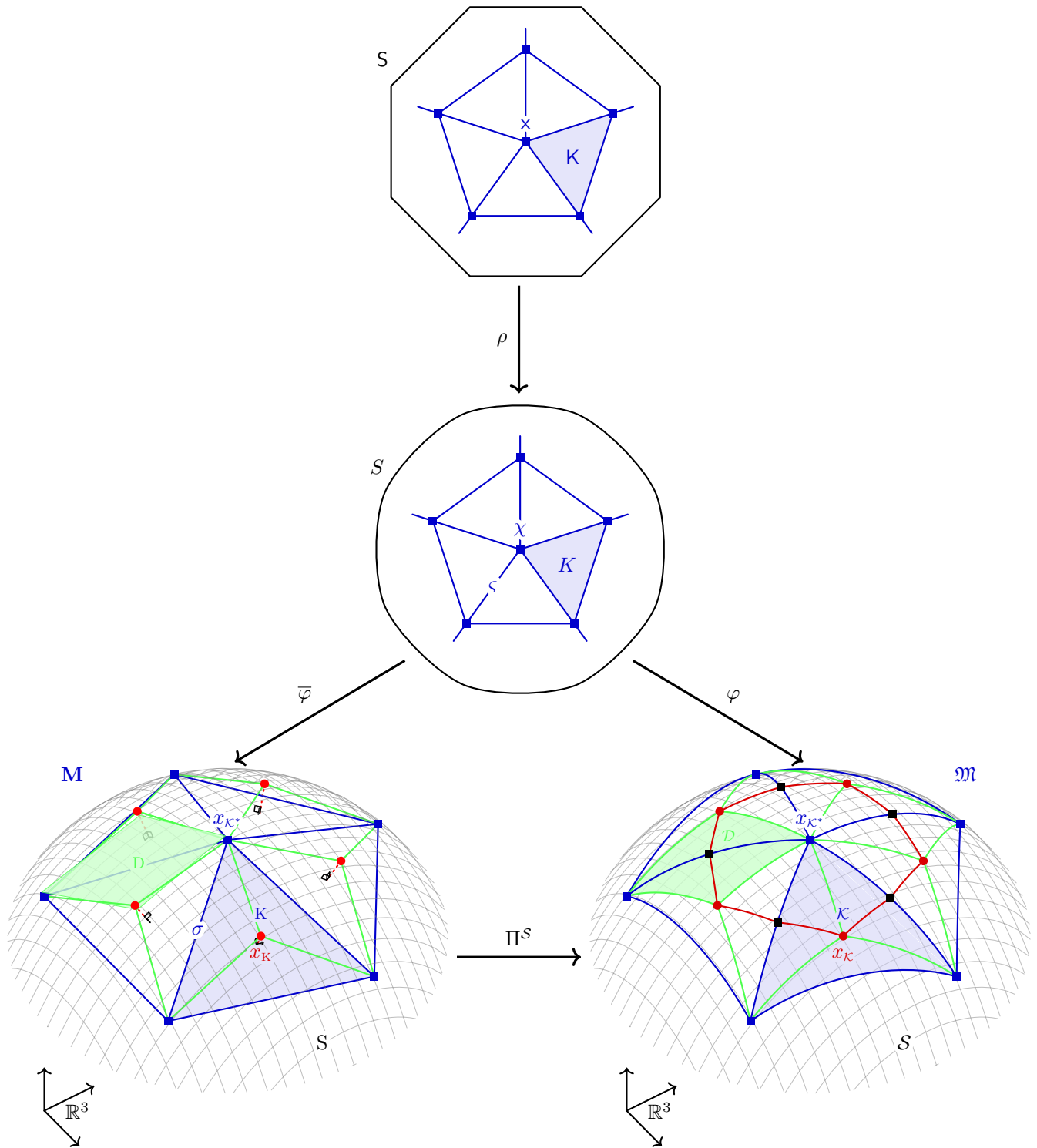


Figure I.4. Triangulated surface from the point of view of the differential geometry

The finite volume mesh is constructed by the barycentric subdivision of $\mathcal{S}$. The finite volume (or covolume) $V_i \mathcal{K}^* \in \mathfrak{M}^*$ around a vertex X_i $x_{\mathcal{K}^*}$ is a region bounded by the piecewise linear curve joining the barycenters of the neighbouring triangles and the midpoints of the edges leading to the neighbouring vertices $X_{i,p}$, $p \in \llbracket 1; m \rrbracket$ where m is the number of neighbours.

In the same way, for a dual control volume $\mathcal{K}^* \in \mathfrak{M}^*$, we set

- $m(\mathcal{K}^*)$, the measure of $\mathcal{K}^*$;
- $\mathfrak{E}_{\mathcal{K}^*}$, the set of edges of $\mathcal{K}^*$;
- $\mathfrak{D}_{\mathcal{K}^*} = \{\mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D} \mid \sigma^* \in \mathfrak{E}_{\mathcal{K}^*}\}$;
- $\nu_{\mathcal{K}^*}$, the outward unit normal vector to $\partial \mathcal{K}^*$;
- $d_{\mathcal{K}^*}$, the diameter of $\mathcal{K}^*$.

I.2.3 Diamond mesh

Next, we construct a diamond set as follows. We consider a couple $(\sigma, \sigma^*) \in \mathfrak{E} \times \mathfrak{E}^*$ such that $\sigma = \mathcal{K} \mid \mathcal{L}$ and $\sigma^* = [x_{\mathcal{K}}, x_{\mathcal{L}}]$. We denote by $x_{\mathcal{K}^*}$ and $x_{\mathcal{L}^*}$ the endpoints of the edge σ . We consider the plane P_{σ, σ^*} passing through the midpoints of the segments $[x_{\mathcal{K}}, x_{\mathcal{K}^*}]$, $[x_{\mathcal{K}}, x_{\mathcal{L}^*}]$, $[x_{\mathcal{L}}, x_{\mathcal{L}^*}]$ and $[x_{\mathcal{L}}, x_{\mathcal{K}^*}]$ (I.8a sketched in black). We then define the quadrilateral diamond cell $\mathcal{D}_{\sigma, \sigma^*}$ whose vertices are the projections of the points $x_{\mathcal{K}}$, $x_{\mathcal{L}}$, $x_{\mathcal{K}^*}$ and $x_{\mathcal{L}^*}$ onto the plane P_{σ, σ^*} in the direction of the vector $\sigma \times \sigma^*$ (I.8a). By abuse of notation, we still denote by σ and σ^* the diagonals of the diamond cell $\mathcal{D}_{\sigma, \sigma^*}$, as shown on I.8b. A diamond is also illustrated on I.4, bottom left. Notice that the diamond cells are the union of two disjoint triangles and can be non convex. The set of diamond cells is denoted by $\mathfrak{D}$. Our diamond set is a generalization of 2D planar diamond set used in DDFV method developed in [DO05] to 2D surfaces in $\mathbb{R}^3$.

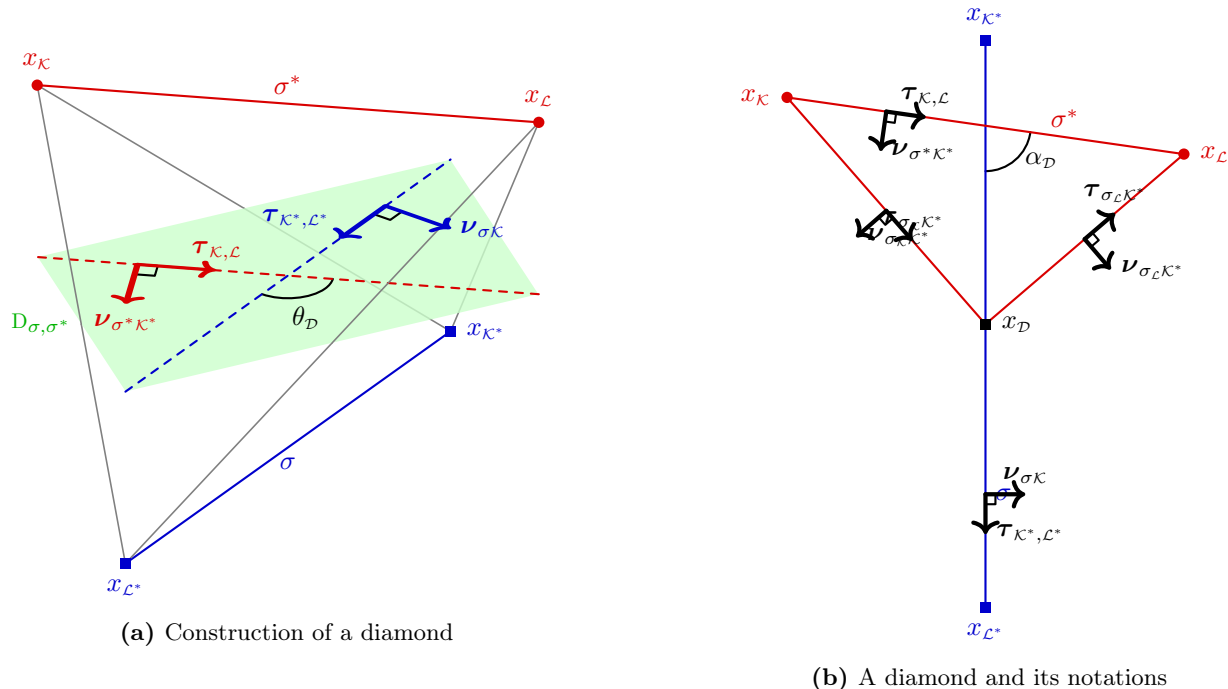
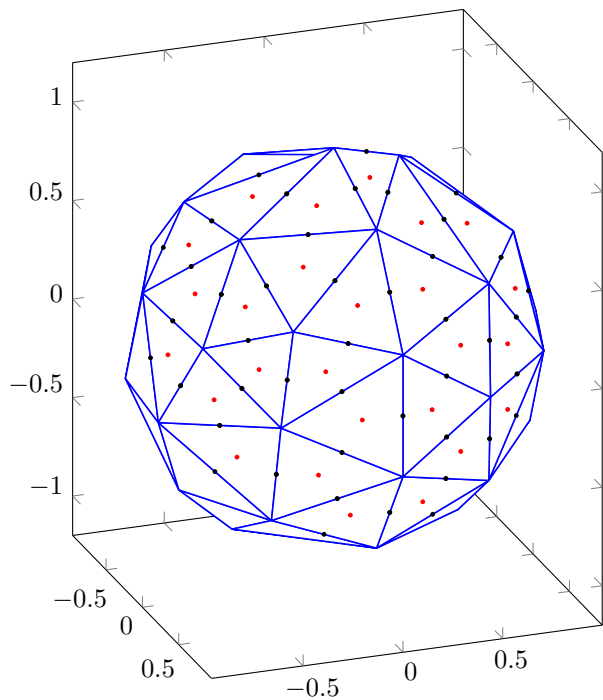
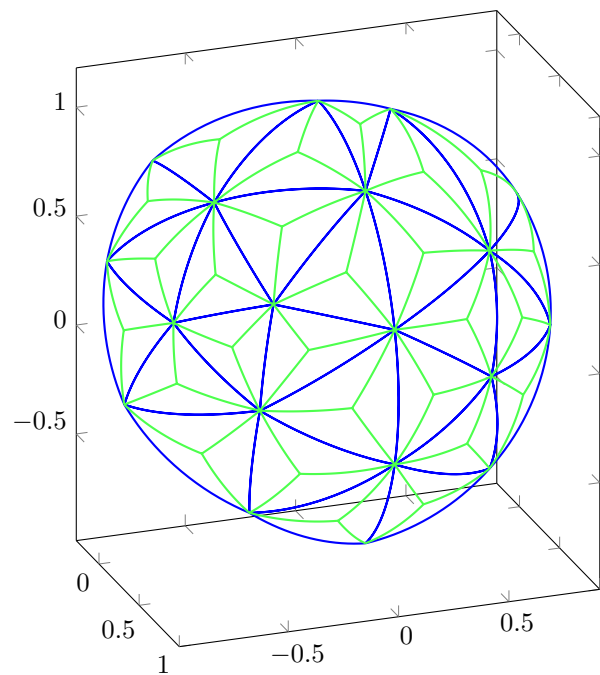


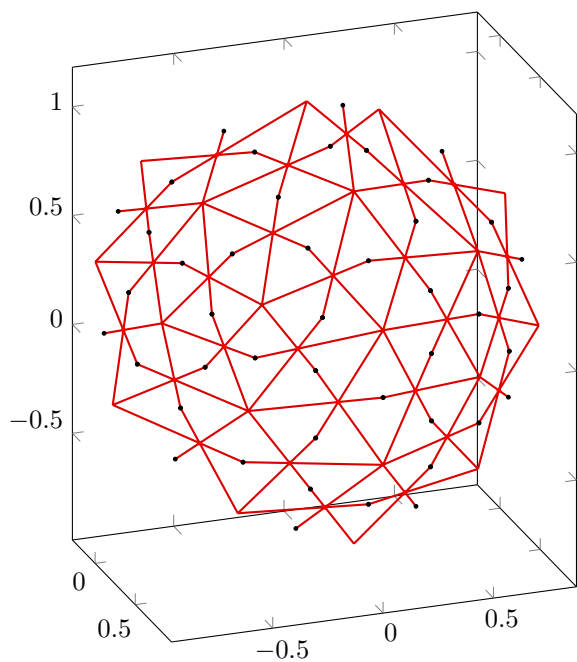
Figure I.8. Diamond



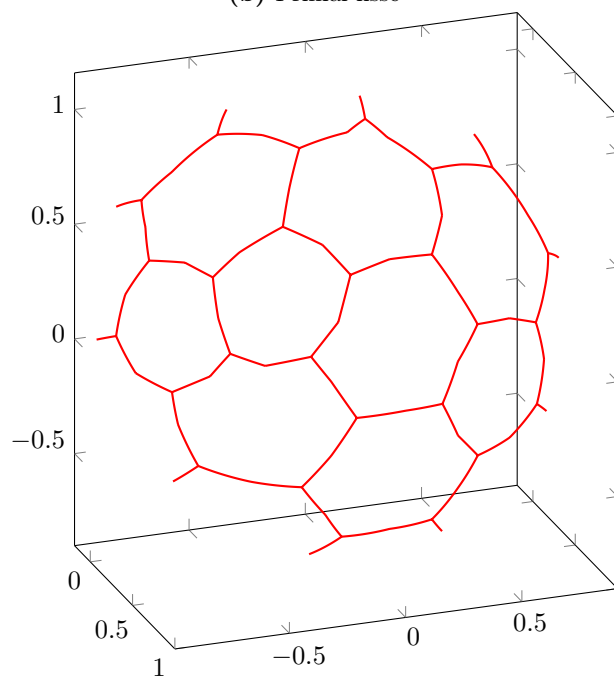
(a) Primal polyédrique



(b) Primal lisse



(c) Dual polyédrique



(d) Dual lisse

Figure I.6

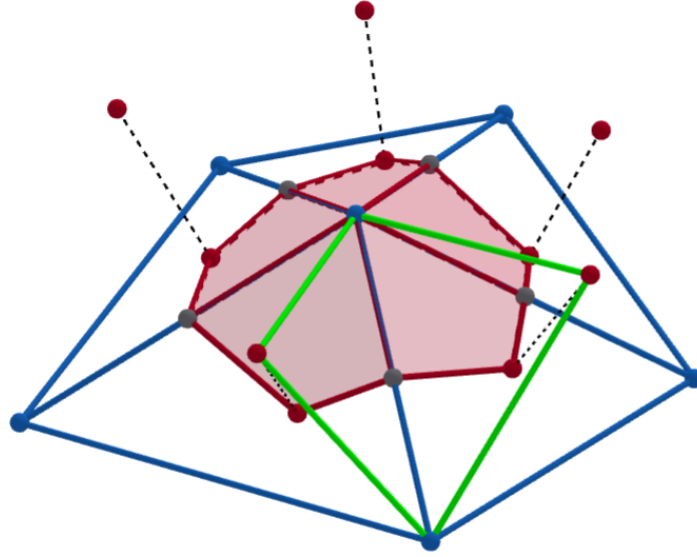


Figure I.9

Remark I.2.1. The plane P_{σ, σ^*} approximates the tangent plane to the surface $\mathcal{S}$ at $?$.

Remark I.2.2. As shown on fig. I.14, the set of diamond cells $\mathfrak{D}$ does not constitute a mesh of the polyhedral surface $\mathcal{S}$.

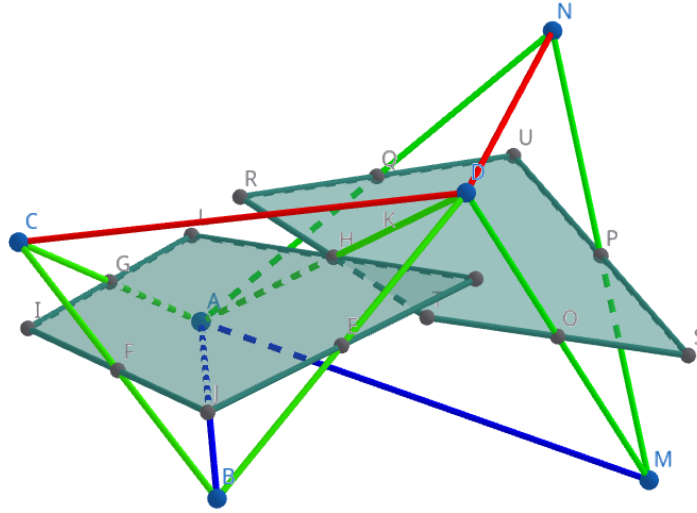


Figure I.10. Adjacent diamonds

Proposition I.2.3 (Mesure d'un diamant plat).

$$m(\mathcal{D}) = \frac{1}{2} \|\sigma \times \sigma^*\| = \frac{1}{2} m(\sigma) m(\sigma^*) \sin(\theta_{\mathcal{D}})$$

For a diamond cell $\mathcal{D}_{\sigma, \sigma^*}$, recall that $(x_{\mathcal{K}}, x_{\mathcal{K}^*}, x_{\mathcal{L}}, x_{\mathcal{L}^*})$ are the vertices of $\mathcal{D}_{\sigma, \sigma^*}$ and note (fig. I.12)

- x_{σ} le milieu de l'arête σ ,
- $x_{\mathcal{D}} = x_{\sigma}$ le centre du diamant $\mathcal{D}$
- $m(\sigma)$, the length of σ , $m(\sigma^*)$ the length of σ^* and $m(\mathcal{D})$ the measure of the diamond cell;
- $\tau_{\mathcal{K}^*, \mathcal{L}^*}$, the unit vector parallel to σ , oriented from $x_{\mathcal{K}^*}$ to $x_{\mathcal{L}^*}$;

- $\nu_{\sigma\kappa}$, the unit vector normal to σ , oriented from x_κ to $x_\mathcal{L}$ (**sortante de κ**);
- $\tau_{\kappa,\mathcal{L}}$, the unit vector parallel to σ^* , oriented from x_κ to $x_\mathcal{L}$;
- $\nu_{\sigma^*\kappa^*}$, the unit vector normal to σ^* , oriented from x_{κ^*} to $x_{\mathcal{L}^*}$ (**sortante de κ^***);
- $\theta_{\mathcal{D}}$, the angle between $\tau_{\kappa^*,\mathcal{L}^*}$ and $\tau_{\kappa,\mathcal{L}}$ (**entre σ et σ^***);
- $d_{\mathcal{D}}$, the diameter of $\mathcal{D}_{\sigma,\sigma^*}$.

Cas barycentrique [Kre10, p. 35]. Le maillage dual barycentrique nécessite de nouvelles notations (voir la **figure**), pour un diamant $\mathcal{D} = \mathcal{D}_{\sigma,\sigma^*}$

Proposition I.2.4 ([Kre10]). On introduit pour chaque diamant les deux bases orthonormées directes $(\tau_{\kappa^*,\mathcal{L}^*}, \nu_{\sigma\kappa})$ et $(\nu_{\sigma^*\kappa^*}, \tau_{\kappa,\mathcal{L}})$. On a les relations

$$\tau_{\kappa,\mathcal{L}} \cdot \nu_{\sigma^*\kappa^*} = \tau_{\kappa^*,\mathcal{L}^*} \cdot \nu_{\sigma\kappa} = 0, \quad (\text{I.2.1})$$

$$\tau_{\kappa^*,\mathcal{L}^*} \cdot \nu_{\sigma^*\kappa^*} = \tau_{\kappa,\mathcal{L}} \cdot \nu_{\sigma\kappa} = \sin(\theta_{\mathcal{D}}), \quad (\text{I.2.2})$$

$$\tau_{\kappa^*,\mathcal{L}^*} \cdot \tau_{\kappa,\mathcal{L}} = \nu_{\sigma^*\kappa^*} \cdot \nu_{\sigma\kappa} = \cos(\theta_{\mathcal{D}}). \quad (\text{I.2.3})$$

I.2.4 DDFV mesh

We call $\mathcal{T}$ a double $(\mathfrak{M}, \mathfrak{M}^*)$ of meshes on the smooth surface $\mathcal{S}$, where $\mathfrak{M}$ is the *primal mesh* and $\mathfrak{M}^*$ is the *dual mesh*. Even though more general situations can be handled, we concentrate in this paper to meshes satisfying the following assumption.

Main assumption. We assume either that all the primal control volumes $\kappa \in \mathfrak{M}$ are star-shaped with respect to x_κ either that all the dual control volumes $\kappa^* \in \mathfrak{M}^*$ are star-shaped with respect to x_{κ^*} .

Remark I.2.5. This hypothesis is not so restrictive and is fulfilled for instance in the case of a primal Voronoï mesh associated to the vertices x_{κ^*} , or in the dual case of a Delaunay conformal triangulation of the domain.

Proposition I.2.6.

$$\mathcal{S} = \bigcup_{\kappa \in \mathfrak{M}} \kappa = \bigcup_{\kappa^* \in \mathfrak{M}^*} \kappa^* = \bigcup_{\mathcal{D} \in \mathfrak{D}} \mathcal{D}$$

I.3 Unknowns / Espaces d'approximation sur les maillages DDFV

The finite volume method associates to all primal control volumes $\kappa \in \mathfrak{M}$, an unknown value u_κ and to all dual control volumes $\kappa^* \in \mathfrak{M}^*$, an unknown value u_{κ^*} . We denote the approximate solution on the mesh $\mathcal{T}$ by

$$u^{\mathcal{T}} = (u^{\mathfrak{M}}, u^{\mathfrak{M}^*}) \quad \text{with} \quad u^{\mathfrak{M}} = (u_\kappa)_{\kappa \in \mathfrak{M}}, \quad u^{\mathfrak{M}^*} = (u_{\kappa^*})_{\kappa^* \in \mathfrak{M}^*}. \quad (\text{I.3.1})$$

On a environ deux fois plus d'inconnues que dans la méthode VF4 mais cela permet de définir une approximation complète du gradient et pas uniquement dans la direction normale, ceci rend la méthode beaucoup plus robuste.

The approximation spaces of all discrete functions $u^{\mathfrak{M}}$, $u^{\mathfrak{M}^*}$ and $u^{\mathcal{T}}$ respectively in the sense of definition I.3.1 is denoted by $\mathbb{R}^{\mathfrak{M}}$, $\mathbb{R}^{\mathfrak{M}^*}$ and $\mathbb{R}^{\mathcal{T}}$ respectively. By introducing the projector

$$\Pi_{\mathcal{T}}^{\mathcal{S}}: \mathbb{R}^{\mathcal{T}} \longrightarrow L^2(\mathcal{S})$$

$$u^{\mathcal{T}} \longmapsto \frac{1}{2} \sum_{\kappa \in \mathfrak{M}} \mathbf{1}_\kappa u_\kappa + \frac{1}{2} \sum_{\kappa^* \in \mathfrak{M}^*} \mathbf{1}_{\kappa^*} u_{\kappa^*}, \quad (\text{I.3.2})$$

we associate to the discrete function $u^{\mathcal{T}}$ the piecewise constant function $\Pi_{\mathcal{T}}^{\mathcal{S}} u^{\mathcal{T}}$. https://theses.hal.science/tel-03265697/file/HDR_VersionImprimee.pdf Avec cette définition, nous utilisons simultanément les valeurs sur le maillage primal et les valeurs sur le maillage dual. En effet, nous avons $u_h = \frac{1}{2} (u_{h,\mathfrak{M}} + u_{h,\mathfrak{M}^*})$, où $u_{h,\mathfrak{M}}$ et $u_{h,\mathfrak{M}^*}$ sont deux reconstructions différentes basées soit sur les valeurs primales soit sur les valeurs duales. L'espace des solutions approchées est dénoté par $\mathbb{H}_{\mathcal{T}}$:

$$\mathbb{H}_{\mathcal{T}} = \left\{ u_h \in L^1(\mathcal{S}) \mid \exists u^{\mathcal{T}} \in \mathbb{R}^{\mathcal{T}} \text{ tel que } u_h = \Pi_{\mathcal{T}}^{\mathcal{S}} u^{\mathcal{T}} \right\}.$$

As a consequence, one can define for any $r \in [1; +\infty]$ the L^r norm of $\Pi_T^S u^T$.

Figure I.11. Fonction constante par mailles primales et duales

On définit également $\xi^{\mathcal{D}} \in (\mathbb{R}^2)^{\mathcal{D}}$ par $\xi^{\mathcal{D}} = (\xi^{\mathcal{D}})_{\mathcal{D} \in \mathcal{D}}$, avec $\xi^{\mathcal{D}} \in \mathbb{R}^2$. In the same way as above, by introducing the projector

$$\Pi_{\mathcal{D}}^S: \begin{aligned} (\mathbb{R}^2)^{\mathcal{D}} &\longrightarrow L^2(\mathcal{M})^2 \\ \xi^{\mathcal{D}} &\longmapsto \sum_{\mathcal{D} \in \mathcal{D}} \mathbf{1}_{\mathcal{D}} \xi^{\mathcal{D}} \end{aligned} \quad (\text{I.3.3})$$

we associate to the discrete function $\xi^{\mathcal{D}}$ the piecewise constant function $\Pi_{\mathcal{D}}^S \xi^{\mathcal{D}}$.

I.4 Construction of the discrete operators

[DDO07, Sec.3] In this section, we approach the gradient, divergence and curl operators by discrete counterparts. We would like to stress that in two dimensions, a distinction is usually made between the vector curl operator from $\mathbb{R}$ to $\mathbb{R}^2$, defined by $\mathbf{rot} u^T = \left(\frac{\partial u^T}{\partial y}, -\frac{\partial u^T}{\partial x} \right)^T$ and the scalar curl operator from $\mathbb{R}^2$ to $\mathbb{R}$, defined by $\text{rot} \xi^{\mathcal{D}} = \frac{\partial \xi_y^{\mathcal{D}}}{\partial x} - \frac{\partial \xi_x^{\mathcal{D}}}{\partial y}$.

Figure I.12 shows the stencils of the different operators and of their combinations. The stencil for the discrete gradient and vector curl operators simply consists of the four **corners** of the diamond-cell $\mathcal{D}$. The stencil for the discrete divergence and scalar curl operators consists of the diamonds associated to the edges of the primal and dual cells. Arrows are displayed on I.12 to represent the normal and tangential components of the vector fields associated to the diamonds. The stencils for the discrete laplacien on the primal and dual cells respectively consist of the blue and red circles on the figure.

I.4.1 Construction of the discrete gradient and vector curl operators on the diamond cells

We define the discrete gradient of a function u by its values on the **diamond-cells of the mesh**. We follow [CVV99; DO05] and compute the mean-value of the gradient of any function u on such a cell $\mathcal{D}$ by the following formula:

$$\mathbf{m}(\mathcal{D}) \langle \nabla u|_{\mathcal{D}} \rangle = \int_{\mathcal{D}} \nabla u(x) \, dx = \int_{\partial \mathcal{D}} u(\xi) \mathbf{n}(\xi) \, d\xi = \sum_{\sigma \in \partial \mathcal{D}} \int_{\sigma} u(\xi) \mathbf{n}(\xi) \, d\xi, \quad (\text{I.4.1})$$

where $\mathbf{n}(\xi)$ stands for the outward unit normal vector to $\mathcal{D}$ at point ξ . The integrals in I.4.1 can be approximated by the following formula:

$$\int_{[x_K, x_{K^*}]} u(\xi) \, d\xi \approx \ell_{[x_K, x_{K^*}]} \frac{u(x_K) + u(x_{K^*})}{2},$$

where $\ell_{[x_K, x_{K^*}]}$ denotes the length of the segment $[x_K, x_{K^*}]$. Summing the contributions of the different vertices of $\mathcal{D}$ and using **elementary geometrical equalities** allows us to give the *definition* of the discrete gradient $\nabla^{\mathcal{D}}$ on $\mathcal{D}$.

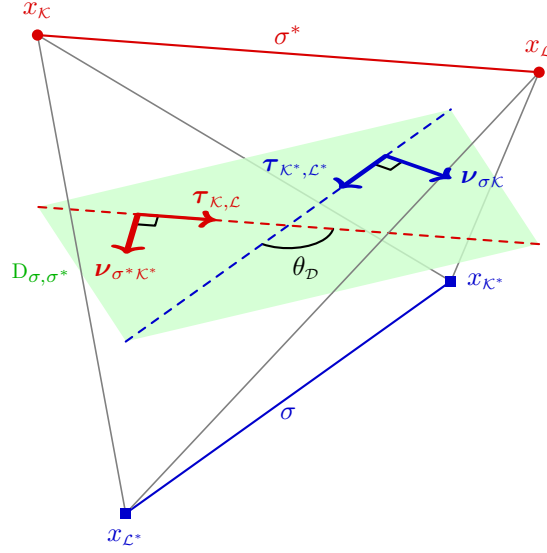


Figure I.12. 3D diamond and its bases / Stencils for the discrete operators

Discrete gradient [ABH07; Kre10]

We consider the discrete gradient introduced by Coudière and *al.* in [CVV99] and applied by Hermeline [Her00] and Domelevo and Omnès [DO05] in the framework of DDFV schemes described above.

Definition I.4.1 (Discrete gradient). The *discrete gradient operator* $\nabla^{\mathfrak{D}} : \mathbb{R}^T \rightarrow (\mathbb{R}^2)^{\mathfrak{D}}$ can be defined as follows: for any $u^T \in \mathbb{R}^T$, $\nabla^{\mathfrak{D}} u^T = (\nabla^{\mathfrak{D}} u^T)_{\mathfrak{D} \in \mathfrak{D}}$ is the function, constant on each diamond cell $\mathfrak{D}_{\sigma, \sigma^*}$, given by

$$\nabla^{\mathfrak{D}} u^T \cdot \tau_{K, L} = \frac{u_L - u_K}{m(\sigma^*)}, \quad \nabla^{\mathfrak{D}} u^T \cdot \tau_{K^*, L^*} = \frac{u_{L^*} - u_{K^*}}{m(\sigma)}, \quad (\text{I.4.2})$$

where $\mathfrak{D}_{\sigma, \sigma^*}$ is noted $\mathfrak{D}$ when no confusion can arise.

Remark I.4.2. Comme il y a une correspondance univoque entre un diamant courbe $\mathfrak{D}$ et un diamant plat $\mathfrak{D}$, on aurait également pu utiliser la notation $\nabla^{\mathfrak{D}}$. Le gradient d'une fonction n'est calculé qu'à partir des longueurs des arêtes et des valeurs de la fonction à leurs extrémités. Computing the discrete gradient only requires the values of u^T at the nodes of the primal and dual meshes.

Remark that $\nabla^{\mathfrak{D}} u^T$ can be expressed in the $(\nu_{\sigma K}, \nu_{\sigma^* K^*})$ basis in the following way:

$$\nabla^{\mathfrak{D}} u^T = \frac{1}{\sin \theta_{\mathfrak{D}}} \left(\frac{u_L - u_K}{m(\sigma^*)} \nu_{\sigma K} + \frac{u_{L^*} - u_{K^*}}{m(\sigma)} \nu_{\sigma^* K^*} \right) \quad (\text{I.4.3})$$

thanks to the following lemma.

Lemma I.4.3. Using the notations and orientation conventions, for any vector $\xi \in \mathbb{R}^2$ we have

$$(\sin \theta_{\mathfrak{D}}) \xi = (\xi, \tau_{K^*, L^*}) \nu_{\sigma^* K^*} + (\xi, \tau_{K, L}) \nu_{\sigma K}.$$

Remark I.4.4. En 2D, d'après la Proposition I.2.3, on peut réécrire le gradient discret sous la forme

$$\nabla^{\mathfrak{D}} u^T = \frac{1}{2m(\mathfrak{D})} \left[(u_L - u_K) m(\sigma) \nu + (u_{L^*} - u_{K^*}) m(\sigma^*) \nu^* \right]. \quad (\text{I.4.4})$$

Note that formula I.4.4 is exact for polynomials of degree one.

Explicitons l'écriture de l'opérateur gradient discret sous forme d'une matrice $\mathbf{G}_{\mathfrak{D}} \in \mathbb{R}^{2\mathfrak{D} \times T}$: ...

In the same way, we may approach the vector curl operator $\mathbf{rot} \bullet = \left(\frac{\partial \bullet}{\partial y}, -\frac{\partial \bullet}{\partial x} \right)^{\top}$ by a discrete vector curl operator.

Discrete vector curl operator (Gradient tourné)

« Le gradient tourné est le tourné du gradient »

$$\int \nabla u \cdot \boldsymbol{\tau} \, ds = u(1) - u(0)$$

$$\int \nabla^\perp u \cdot \boldsymbol{\nu} \, ds = u(1) - u(0)$$

Definition I.4.5 (Discrete vector curl operator). The *discrete vector curl operator* $\mathbf{rot}^\mathfrak{D} : \mathbb{R}^\mathcal{T} \rightarrow (\mathbb{R}^2)^\mathfrak{D}$ can be defined as follows: for any $u^\mathcal{T} \in \mathbb{R}^\mathcal{T}$, $\mathbf{rot}^\mathfrak{D} u^\mathcal{T} = (\mathbf{rot}^\mathfrak{D} u^\mathcal{T})_{\mathfrak{D} \in \mathfrak{D}}$ is the function, constant on each diamond cell $\mathfrak{D}_{\sigma, \sigma^*}$, given by

$$(\mathbf{rot}^\mathfrak{D} u^\mathcal{T}, \boldsymbol{\nu}_{\sigma^* \kappa^*}) = -\frac{u_\mathcal{L} - u_\mathcal{K}}{m(\sigma^*)}, \quad (\mathbf{rot}^\mathfrak{D} u^\mathcal{T}, \boldsymbol{\nu}_{\sigma \kappa}) = -\frac{u_{\mathcal{L}^*} - u_{\mathcal{K}^*}}{m(\sigma)} \quad (\text{I.4.5})$$

where the unit vectors $\boldsymbol{\tau}$ and $\boldsymbol{\tau}^*$ are such that $(\boldsymbol{\nu}, \boldsymbol{\tau})$ and $(\boldsymbol{\nu}^*, \boldsymbol{\tau}^*)$ are orthonormal positively oriented bases of $\mathbb{R}^2$.

Remark that $\mathbf{rot}^\mathfrak{D} u^\mathcal{T}$ can be expressed in the $(\boldsymbol{\tau}_{\mathcal{K}^*, \mathcal{L}^*}, \boldsymbol{\tau}_{\mathcal{K}, \mathcal{L}})$ basis in the following way:

$$\mathbf{rot}^\mathfrak{D} u^\mathcal{T} = -\frac{1}{\sin \theta_\mathfrak{D}} \left(\frac{u_\mathcal{L} - u_\mathcal{K}}{m(\sigma^*)} \boldsymbol{\tau}_{\mathcal{K}^*, \mathcal{L}^*} + \frac{u_{\mathcal{L}^*} - u_{\mathcal{K}^*}}{m(\sigma)} \boldsymbol{\tau}_{\mathcal{K}, \mathcal{L}} \right). \quad (\text{I.4.6})$$

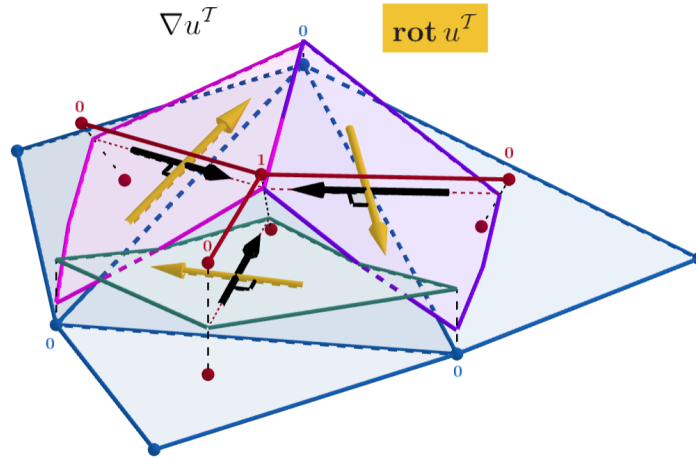


Figure I.13

Remark I.4.6 ([DDO07, Remark 3.3]). In a connected domain, the discrete gradient and vector curl of a given $u^\mathcal{T} \in \mathbb{R}^\mathcal{T}$ vanish if and only if there exist two constants $c^\mathfrak{M}$ and $c^{\mathfrak{M}^*}$, such that

$$\forall \mathcal{K} \in \mathfrak{M}, \quad u_\mathcal{K} = c^\mathfrak{M}, \quad \forall \mathcal{K}^* \in \mathfrak{M}^*, \quad u_{\mathcal{K}^*} = c^{\mathfrak{M}^*}.$$

The fact that $c^\mathfrak{M}$ and $c^{\mathfrak{M}^*}$ may differ one from the other means that such a $u^\mathcal{T}$ may in general present oscillations.

I.4.2 Produits scalaires

Section I.3 leads us to the definition of the following discrete scalar products. We denote by $(\cdot, \cdot)_\mathcal{T}$ the inner product on $\mathbb{R}^\mathcal{T}$ given by

$$(u^\mathcal{T}, v^\mathcal{T})_\mathcal{T} \stackrel{\text{d\'ef.}}{=} \sum_{\mathcal{K} \in \mathfrak{M}} \mu_\mathcal{K} u_\mathcal{K} v_\mathcal{K} + \sum_{\mathcal{K}^* \in \mathfrak{M}^*} \mu_{\mathcal{K}^*} u_{\mathcal{K}^*} v_{\mathcal{K}^*}, \quad (\text{I.4.7})$$

which stands for a discrete L^2 inner product. In the same way, we define a discrete scalar product on the diamond mesh for all $\xi^\mathfrak{D}, \phi^\mathfrak{D} \in (\mathbb{R}^2)^\mathfrak{D}$

$$(\xi^\mathfrak{D}, \phi^\mathfrak{D})_\mathfrak{D} \stackrel{\text{d\'ef.}}{=} \sum_{\mathfrak{D} \in \mathfrak{D}} \mu_\mathfrak{D} \xi^\mathfrak{D} \cdot \phi^\mathfrak{D}. \quad (\text{I.4.8})$$

Pour l'instant, nous ne précisons pas la valeur des poids μ_κ , μ_{κ^*} et $\mu_\mathcal{D}$. Nous y reviendrons ultérieurement. Les deux normes correspondantes sont notées par $\|\cdot\|_{2,\mathcal{T}}$ et $\|\cdot\|_{2,\mathcal{D}}$. Plus généralement, nous posons pour tout $u^\mathcal{T} \in \mathbb{R}^\mathcal{T}$, $\xi^\mathcal{D} \in (\mathbb{R}^2)^\mathcal{D}$ et $1 \leq p < +\infty$:

$$\|u^\mathcal{T}\|_{p,\mathcal{T}} \stackrel{\text{déf.}}{=} \left(\frac{1}{2} \sum_{\kappa \in \mathfrak{M}} \mu_\kappa |u_\kappa|^p + \frac{1}{2} \sum_{\kappa^* \in \mathfrak{M}^*} \mu_{\kappa^*} |u_{\kappa^*}|^p \right)^{1/p}, \quad \|\xi^\mathcal{D}\|_{p,\mathcal{D}} \stackrel{\text{déf.}}{=} \left(\sum_{\mathcal{D} \in \mathcal{D}} \mu_\mathcal{D} |\xi^\mathcal{D}|^p \right)^{1/p}, \quad (\text{I.4.9})$$

$$\|u^\mathcal{T}\|_{\infty,\mathcal{T}} \stackrel{\text{déf.}}{=} \max \left(\max_{\kappa \in \mathfrak{M}} |u_\kappa|, \max_{\kappa^* \in \mathfrak{M}^*} |u_{\kappa^*}| \right), \quad \|\xi^\mathcal{D}\|_{\infty,\mathcal{D}} \stackrel{\text{déf.}}{=} \max_{\mathcal{D} \in \mathcal{D}} |\xi^\mathcal{D}|. \quad (\text{I.4.10})$$

Par exemple,

$$\|\nabla^\mathcal{D} u^\mathfrak{M}\|_{2,\mathcal{D}}^2 = \sum_{\mathcal{D} \in \mathcal{D}} \mu_\mathcal{D} |\nabla^\mathcal{D} u^\mathfrak{M}|^2.$$

Calcul des poids μ_κ et μ_{κ^*}

Following Remark I.4.6, nous nous placerons dans l'espace des fonctions orthogonales aux gradients nuls, à savoir

$$\tilde{\mathbb{R}}^\mathcal{T} \stackrel{\text{déf.}}{=} \mathbb{R}^\mathcal{T} / \mathbb{R} = \left\{ u^\mathcal{T} \in \mathbb{R}^\mathcal{T} \mid \sum_{\kappa \in \mathfrak{M}} \mu_\kappa u_\kappa = \sum_{\kappa^* \in \mathfrak{M}^*} \mu_{\kappa^*} u_{\kappa^*} = 0 \right\}. \quad (\text{I.4.11})$$

On prend en général les valeurs

$$\mu_\kappa = \frac{\mathfrak{m}(\mathcal{K})}{2} \quad \text{et} \quad \mu_{\kappa^*} = \frac{\mathfrak{m}(\mathcal{K}^*)}{2}. \quad (\text{I.4.12})$$

On peut prendre d'autres découpages : $\mu_\kappa = \alpha \mathfrak{m}(\mathcal{K})$ et $\mu_{\kappa^*} = (1 - \alpha) \mathfrak{m}(\mathcal{K}^*)$ pour $0 < \alpha < 1$. L'essentiel est d'avoir

$$\|\mathbf{1}\|_{2,\mathcal{T}} = \mathfrak{m}(\mathcal{S}).$$

Calcul des poids $\mu_\mathcal{D}$

Nous avons introduit un produit scalaire Equation (I.4.8) sur les fonctions de $(\mathbb{R}^2)^\mathcal{D}$ de la forme

$$\forall \xi^\mathcal{D}, \phi^\mathcal{D} \in (\mathbb{R}^2)^\mathcal{D}, \quad (\xi^\mathcal{D}, \phi^\mathcal{D})_\mathcal{D} = \sum_{\mathcal{D} \in \mathcal{D}} \mu_\mathcal{D} \xi^\mathcal{D} \cdot \phi^\mathcal{D},$$

où $(\mu_\mathcal{D})_{\mathcal{D} \in \mathcal{D}}$ sont des poids à déterminer. On souhaite que ce produit scalaire soit tel qu'on ait un analogue discret de $\mathbf{rot} \perp \nabla$. En effet, dans le cas continu, par intégration par parties sur la surface sans bord $\mathcal{S}$ on a

$$\forall u, v \in H^1(\mathcal{S}), \quad \int_{\mathcal{S}} \nabla u \cdot \nabla^\perp v \, ds = 0.$$

Le but de cette partie est d'obtenir l'analogue discret de cette formule. Calculons alors les poids $(\mu_\mathcal{D})_{\mathcal{D} \in \mathcal{D}}$ de telle sorte que pour tout $u^\mathcal{T}, v^\mathcal{T} \in \mathbb{R}^\mathcal{T}$, on ait $(\mathbf{rot}^\mathcal{D} u^\mathcal{T}, \nabla^\mathcal{D} v^\mathcal{T})_\mathcal{D} = 0$. D'après les expressions Equations (I.4.3) and (I.4.6) et la Proposition I.2.4, on a

$$(\mathbf{rot}^\mathcal{D} u^\mathcal{T}, \nabla^\mathcal{D} v^\mathcal{T})_\mathcal{D} = - \sum_{\mathcal{D} \in \mathcal{D}} \frac{\mu_\mathcal{D}}{\sin(\theta_\mathcal{D}) \mathfrak{m}(\sigma) \mathfrak{m}(\sigma^*)} \psi_\mathcal{D}(u^\mathcal{T}, v^\mathcal{T})$$

avec $\psi_\mathcal{D}(u^\mathcal{T}, v^\mathcal{T}) = (u_\mathcal{L} - u_\mathcal{K})(v_{\mathcal{L}^*} - v_{\mathcal{K}^*}) + (u_{\mathcal{L}^*} - u_{\mathcal{K}^*})(v_\mathcal{L} - v_\mathcal{K})$.

Toute fonction $u^\mathcal{T} \in \mathbb{R}^\mathcal{T}$, peut s'écrire sous la forme $u^\mathcal{T} = \sum_{\kappa \in \mathfrak{M}} \mathbf{1}^\mathcal{T}(\kappa) u_\kappa + \sum_{\kappa^* \in \mathfrak{M}^*} \mathbf{1}^\mathcal{T}(\kappa^*) u_{\kappa^*}$. Il y a donc quatre cas à traiter; $(u^\mathcal{T}, v^\mathcal{T}) = (\mathbf{1}^\mathcal{T}(\kappa), \mathbf{1}^\mathcal{T}(\kappa))$, $(u^\mathcal{T}, v^\mathcal{T}) = (\mathbf{1}^\mathcal{T}(\kappa^*), \mathbf{1}^\mathcal{T}(\kappa^*))$, $(u^\mathcal{T}, v^\mathcal{T}) = (\mathbf{1}^\mathcal{T}(\kappa), \mathbf{1}^\mathcal{T}(\kappa^*))$ et $(u^\mathcal{T}, v^\mathcal{T}) = (\mathbf{1}^\mathcal{T}(\kappa^*), \mathbf{1}^\mathcal{T}(\kappa))$. Dans les deux premiers cas, $\psi_\mathcal{D}(u^\mathcal{T}, v^\mathcal{T}) = 0$. Étudions les deux derniers cas pour deux mailles $\kappa \in \mathfrak{M}$ et $\kappa^* \in \mathfrak{M}^*$ quelconques. On pose $u^\mathcal{T} = \mathbf{1}^\mathcal{T}(\kappa)$ et $v^\mathcal{T} = \mathbf{1}^\mathcal{T}(\kappa^*)$. Si $\mathcal{D}_\kappa \cap \mathcal{D}_{\kappa^*} = \emptyset$, alors $\psi_\mathcal{D}(u^\mathcal{T}, v^\mathcal{T}) = 0$. Sinon, les mailles κ et κ^* ont exactement deux diamants en commun et on note $\mathcal{D}_\kappa \cap \mathcal{D}_{\kappa^*} = \{\mathcal{D}_1, \mathcal{D}_2\}$. On les représente sur la Figure I.14. À nouveau, d'après les expressions Equations (I.4.3) and (I.4.6), on a

$$\mathbf{rot}^{\mathcal{D}_1} u^\mathcal{T} = \frac{1}{\sin \theta_{\mathcal{D}_1}} \frac{-1}{\mathfrak{m}(\sigma_{\mathcal{D}_1}^*)} \tau_{\kappa^*, \kappa^*_1}, \quad \mathbf{rot}^{\mathcal{D}_2} u^\mathcal{T} = \frac{1}{\sin \theta_{\mathcal{D}_2}} \frac{1}{\mathfrak{m}(\sigma_{\mathcal{D}_2}^*)} \tau_{\kappa^*, \kappa^*_2} \quad (\text{I.4.13})$$

et

$$\nabla^{D_1} v^T = \frac{1}{\sin \theta_{D_1}} \frac{-1}{m(\sigma_1)} \nu_{\sigma^*_1 \kappa^*}, \quad \nabla^{D_2} v^T = \frac{1}{\sin \theta_{D_2}} \frac{-1}{m(\sigma_2)} \nu_{\sigma^*_2 \kappa^*}. \quad (\text{I.4.14})$$

Ainsi,

$$(\text{rot}^{\mathfrak{D}} u^T, \nabla^{\mathfrak{D}} v^T)_{\mathfrak{D}} = \frac{\mu_{D_1}}{\sin(\theta_{D_1}) m(\sigma_1) m(\sigma^*_1)} - \frac{\mu_{D_2}}{\sin(\theta_{D_2}) m(\sigma_2) m(\sigma^*_2)}$$

ce qui impose que pour tout $i \in \{1, 2\}$, $\mu_{D_i} = \sin(\theta_{D_i}) m(\sigma_i) m(\sigma^*_i)$. Le calcul pour le cas $(u^T, v^T) = (\mathbf{1}^T(\kappa^*), \mathbf{1}^T(\kappa))$ donne le même résultat et on décide de multiplier l'expression par $\frac{1}{2}$ de sorte à avoir

$$\boxed{\forall \mathcal{D} \in \mathfrak{D}, \quad \mu_{\mathcal{D}} = m(\mathcal{D}).} \quad (\text{I.4.15})$$

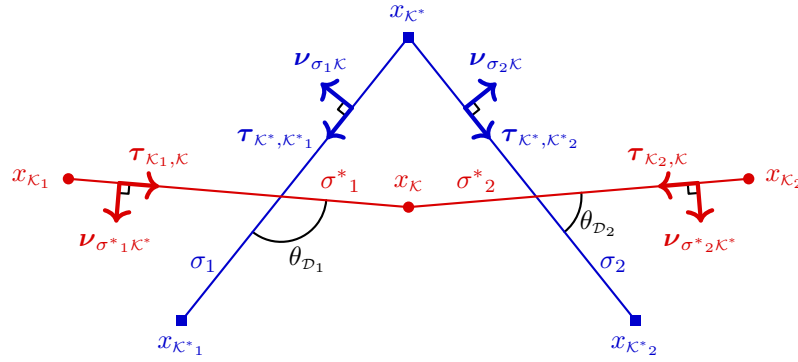


Figure I.14. Diamants voisins

Theorem I.4.7.

$$\forall u^T, v^T \in \mathbb{R}^T, \quad (\text{rot}^{\mathfrak{D}} u^T, \nabla^{\mathfrak{D}} v^T)_{\mathfrak{D}} = 0.$$

Proposition I.4.8. Soient $u^T \in \mathbb{R}^T$ et $\xi^{\mathfrak{D}} \in (\mathbb{R}^2)^{\mathfrak{D}}$.

- Normes discrètes :

$$\begin{aligned} \|u^T\|_{2,T}^2 &= \alpha \sum_{\kappa \in \mathfrak{N}} m(\kappa) u_{\kappa}^2 + (1 - \alpha) \sum_{\kappa^* \in \mathfrak{N}^*} m(\kappa^*) u_{\kappa^*}^2 \\ \|\xi^{\mathfrak{D}}\|_{2,\mathfrak{D}}^2 &= \sum_{\mathcal{D} \in \mathfrak{D}} m(\mathcal{D}) |\xi^{\mathcal{D}}|^2 \end{aligned}$$

- Normes sur la surface lisse $\mathcal{S}$:

$$\begin{aligned} \|\Pi_{\mathcal{T}}^{\mathcal{S}} u^T\|_{L^2(\mathcal{S})}^2 &= \frac{1}{2} \sum_{\kappa \in \mathfrak{N}} m(\kappa) u_{\kappa}^2 + \frac{1}{2} \sum_{\kappa^* \in \mathfrak{N}^*} m(\kappa^*) u_{\kappa^*}^2 \\ \|\Pi_{\mathfrak{D}}^{\mathcal{S}} \xi^{\mathfrak{D}}\|_{L^2(\mathcal{S})^2}^2 &= \sum_{\mathcal{D} \in \mathfrak{D}} m(\mathcal{D}) |\xi^{\mathcal{D}}|^2, \end{aligned}$$

Remark I.4.9. Nous venons de montrer que les expressions des poids μ_{κ} , μ_{κ^*} et $\mu_{\mathcal{D}}$ (Equations (I.4.12) and (I.4.15)) sont proportionnelles à la mesure de leur analogue plat. Aussi, d'un point de vue de l'implémentation numérique, calculer la mesure des mailles plates est préférable à calculer la mesure des mailles courbes. Comme le montre la Proposition I.4.8, les poids associés aux normes discrètes $\|\cdot\|_{2,T}$ et $\|\cdot\|_{2,\mathfrak{D}}$ sont les mesures des mailles plates et les poids associés aux normes sur la surface lisse $\|\Pi_{\mathcal{T}}^{\mathcal{S}} \cdot\|_{L^2(\mathcal{S})}^2$ et $\|\Pi_{\mathfrak{D}}^{\mathcal{S}} \cdot\|_{L^2(\mathcal{S})^2}^2$ sont les mesures des mailles courbes.

I.4.3 Écriture des produits scalaires sous forme matricielle

Conventions. Soient $u^T \in \mathbb{R}^T$ et $\xi^{\mathcal{D}} \in (\mathbb{R}^2)^{\mathcal{D}}$. On note $m = \#\mathfrak{M}$, $m^* = \#\mathfrak{M}^*$ et $d = \#\mathcal{D}$. On désigne par

$$U^T \stackrel{\text{déf.}}{=} (u_{\mathcal{K}_1} \quad \cdots \quad u_{\mathcal{K}_m} \quad u_{\mathcal{K}^*_1} \quad \cdots \quad u_{\mathcal{K}^*_{m^*}}) \quad \text{et} \quad \Xi^{\mathcal{D}} \stackrel{\text{déf.}}{=} (\xi^{\mathcal{D}_1} \quad \cdots \quad \xi^{\mathcal{D}_d})$$

l'écriture vectorielle de u^T et de $\xi^{\mathcal{D}}$.

Le produit scalaire entre deux approximations scalaires u^T et v^T peut s'écrire sous forme matricielle:

$$(u^T, v^T)_T = (U^T)^\top M_T V^T$$

où M_T est la matrice diagonale inversible $M_T = \text{diag}[(\mu_{\mathcal{K}})_{\mathcal{K} \in \mathfrak{M}}, (\mu_{\mathcal{K}^*})_{\mathcal{K}^* \in \mathfrak{M}^*}]$.

Le produit scalaire entre deux approximations vectorielles $\xi^{\mathcal{D}}$ et $\phi^{\mathcal{D}}$ peut s'écrire sous forme matricielle également. On note Σ la base (σ, σ^*) et $\Xi_\Sigma^{\mathcal{D}}$ (resp. $\Phi_\Sigma^{\mathcal{D}}$) l'écriture du vecteur $\xi^{\mathcal{D}}$ (resp. $\phi^{\mathcal{D}}$) dans la base Σ . Par un abus de notation, on écrit $\Xi_\Sigma^{\mathcal{D}} = (\Xi_\Sigma^{\mathcal{D}})_{\mathcal{D} \in \mathcal{D}}$ et de même pour $\Phi_\Sigma^{\mathcal{D}}$. On peut alors écrire

$$(\xi^{\mathcal{D}}, \phi^{\mathcal{D}})_{\mathcal{D}} = \sum_{\mathcal{D} \in \mathcal{D}} (\Xi_\Sigma^{\mathcal{D}})^\top M_\Sigma^{\mathcal{D}} \Phi_\Sigma^{\mathcal{D}}$$

où

$$M_\Sigma^{\mathcal{D}} = \mu_{\mathcal{D}} \begin{pmatrix} m(\sigma)^2 \tau_{\mathcal{K}^*, \mathcal{L}^*} \cdot \tau_{\mathcal{K}^*, \mathcal{L}^*} & m(\sigma) m(\sigma^*) \tau_{\mathcal{K}^*, \mathcal{L}^*} \cdot \tau_{\mathcal{K}, \mathcal{L}} \\ m(\sigma) m(\sigma^*) \tau_{\mathcal{K}^*, \mathcal{L}^*} \cdot \tau_{\mathcal{K}, \mathcal{L}} & m(\sigma^*)^2 \tau_{\mathcal{K}^*, \mathcal{L}^*} \cdot \tau_{\mathcal{K}^*, \mathcal{L}^*} \end{pmatrix} = \mu_{\mathcal{D}} \begin{pmatrix} m(\sigma)^2 & m(\sigma) m(\sigma^*) \cos(\theta_{\mathcal{D}}) \\ m(\sigma) m(\sigma^*) \cos(\theta_{\mathcal{D}}) & m(\sigma^*)^2 \end{pmatrix}.$$

La matrice $\frac{1}{\mu_{\mathcal{D}}} M_\Sigma^{\mathcal{D}}$ est la matrice de Gram du produit scalaire dans la base Σ . On remarque qu'on aurait également pu se placer dans les bases $(\tau_{\mathcal{K}^*, \mathcal{L}^*}, \tau_{\mathcal{K}, \mathcal{L}})$ ou $(\nu_{\sigma \mathcal{K}}, \nu_{\sigma^* \mathcal{K}^*})$.

Remark I.4.10 (HdR Stella). Pour $u^T \in \mathbb{R}^T$ et $\mathcal{D} \in \mathcal{D}$, nous définissons $\delta^{\mathcal{D}} u^T = \begin{pmatrix} u_{\mathcal{K}} - u_{\mathcal{L}} \\ u_{\mathcal{K}^*} - u_{\mathcal{L}^*} \end{pmatrix}$. Puis, pour tout $u^T, v^T \in \mathbb{R}^T$, nous pouvons écrire

$$(\nabla^{\mathcal{D}} u^T, \nabla^{\mathcal{D}} v^T)_{\mathcal{D}} = \sum_{\mathcal{D} \in \mathcal{D}} \delta^{\mathcal{D}} u^T \cdot M_{\mathcal{D}} \delta^{\mathcal{D}} v^T, \quad \text{où} \quad M_{\mathcal{D}} = \frac{1}{4 \mu_{\mathcal{D}}^2} M_\Sigma^{\mathcal{D}} \quad \forall \mathcal{D} \in \mathcal{D}.$$

Finalement, en introduisant la matrice symétrique diagonale par blocs $M_\Sigma^{\mathcal{D}} \stackrel{\text{déf.}}{=} \text{diag}[(M_\Sigma^{\mathcal{D}})_{\mathcal{D} \in \mathcal{D}}]$, on peut écrire le produit scalaire vectoriel sous la forme

$$(\xi^{\mathcal{D}}, \phi^{\mathcal{D}})_{\mathcal{D}} = (\Xi_\Sigma^{\mathcal{D}})^\top M_\Sigma^{\mathcal{D}} \Phi_\Sigma^{\mathcal{D}}. \tag{I.4.16}$$

Dans la suite, nous noterons simplement $M_{\mathcal{D}} = M_\Sigma^{\mathcal{D}}$. Par exemple, le Theorem I.4.7 se réécrit sous la forme

$$\forall u^T, v^T \in \mathbb{R}^T, \quad (U^T)^\top (\mathbf{R}_{\mathcal{D}})^\top M_{\mathcal{D}} \mathbf{G}_{\mathcal{D}} V^T = 0.$$

I.4.4 Construction of the discrete divergence and scalar curl operators on the primal and dual meshes

Next, we choose to define the discrete divergence of a vector field $\xi^{\mathcal{D}}$ by its values both on the primal and dual cells of the mesh. A very natural way to do so on the primal cell $\mathcal{K}$ is to write

...

On définit alors div (resp. rot) comme l'adjoint de ∇ (resp. $\mathbf{rot}$). On a également $\text{div} \mathbf{rot} = 0$ et $\text{rot} \nabla = 0$.

Discrete divergence

Definition I.4.11 (Discrete divergence). L'opérateur de divergence discrète est défini de la manière suivante:

$$(M_T U^T, \mathbf{D}_T \Xi^{\mathcal{D}}) = - (M_{\mathcal{D}} \mathbf{G}_{\mathcal{D}} U^T, \Xi^{\mathcal{D}})$$

$$\mathbf{D}_T \stackrel{\text{déf.}}{=} -M_T^{-1} \mathbf{G}_{\mathcal{D}}^\top M_{\mathcal{D}}$$

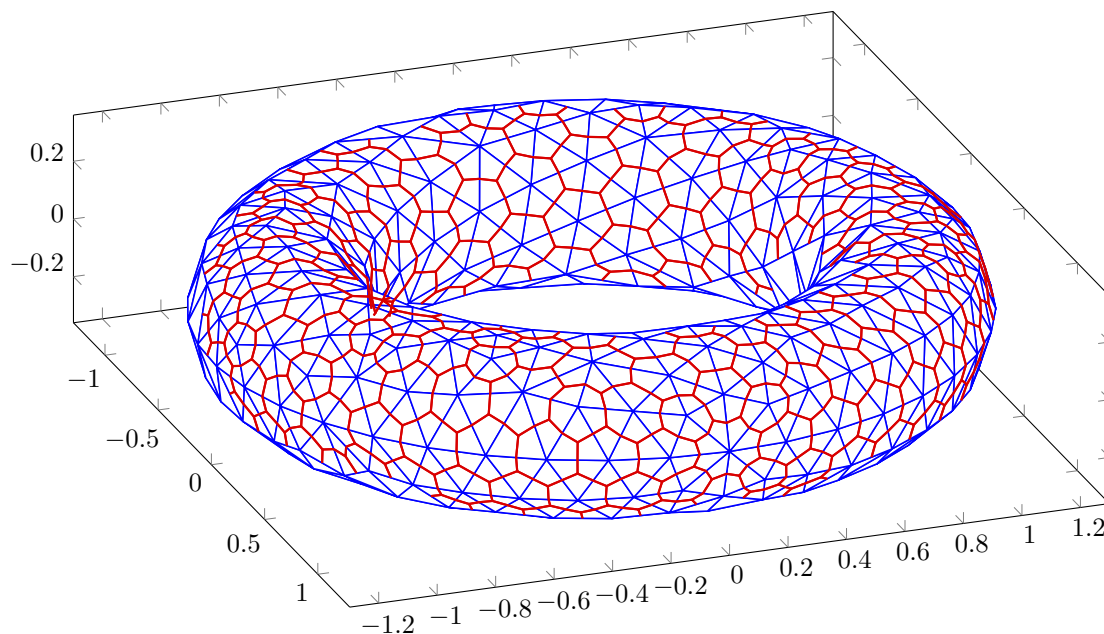


Figure I.7. Polyédrique

scal. vect. scal.

$$\Lambda^0 \begin{array}{c} \xrightarrow{\nabla} \\ \xleftarrow{\text{div}} \end{array} \Lambda^1 \begin{array}{c} \xrightarrow{\text{rot}} \\ \xleftarrow{\nabla^\perp} \end{array} \Lambda^2$$

Figure I.15

Proposition I.4.12. *L'opérateur divergence discrète peut s'écrire sous la forme $\text{div}^T : (\mathbb{R}^2)^\mathfrak{D} \rightarrow \mathbb{R}^T$. Soit $\xi^\mathfrak{D} \in (\mathbb{R}^2)^\mathfrak{D}$, on pose*

$$\text{div}^T \xi^\mathfrak{D} = (\text{div}^\mathfrak{M} \xi^\mathfrak{D}, \text{div}^{\mathfrak{M}^*} \xi^\mathfrak{D}),$$

avec $\text{div}^\mathfrak{M} \xi^\mathfrak{D} = (\text{div}^\mathcal{K} \xi^\mathfrak{D})_{\mathcal{K} \in \mathfrak{M}}$, $\text{div}^{\mathfrak{M}^*} \xi^\mathfrak{D} = (\text{div}^{\mathcal{K}^*} \xi^\mathfrak{D})_{\mathcal{K}^* \in \mathfrak{M}^*}$:

$$\forall \mathcal{K} \in \mathfrak{M}, \quad \text{div}^\mathcal{K} \xi^\mathfrak{D} = \frac{1}{m(\mathcal{K})} \sum_{\mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D}_\mathcal{K}} m(\sigma) \xi^\mathfrak{D} \cdot \boldsymbol{\nu},$$

$$\forall \mathcal{K}^* \in \mathfrak{M}^*, \quad \text{div}^{\mathcal{K}^*} \xi^\mathfrak{D} = \frac{1}{m(\mathcal{K}^*)} \sum_{\mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D}_{\mathcal{K}^*}} m(\sigma^*) \xi^\mathfrak{D} \cdot \boldsymbol{\nu}^*.$$

■ **Proof.**

□

Discrete scalar curl operator

In the same way, we may approach the scalar curl operator $\text{rot} \bullet = \frac{\partial \bullet_y}{\partial x} - \frac{\partial \bullet_x}{\partial y}$ by a discrete scalar curl operator:

Definition I.4.13 (Discrete scalar curl operator). L'opérateur discrete scalar curl est défini de la manière suivante:

$$\mathbf{R}_T \stackrel{\text{déf.}}{=} \mathbf{M}_T^{-1} \mathbf{R}_\mathfrak{D}^\top \mathbf{M}_\mathfrak{D}$$

Proposition I.4.14. $\text{rot}^T : (\mathbb{R}^2)^\mathfrak{D} \rightarrow \mathbb{R}^T$. Soit $\xi^\mathfrak{D} \in (\mathbb{R}^2)^\mathfrak{D}$, on pose

$$\text{rot}^T \xi^\mathfrak{D} = (\text{rot}^\mathfrak{M} \xi^\mathfrak{D}, \text{rot}^{\mathfrak{M}^*} \xi^\mathfrak{D}),$$

with $\text{rot}^\mathfrak{M} \xi^\mathfrak{D} = (\text{rot}^\mathcal{K} \xi^\mathfrak{D})_{\mathcal{K} \in \mathfrak{M}}$, $\text{rot}^{\mathfrak{M}^*} \xi^\mathfrak{D} = (\text{rot}^{\mathcal{K}^*} \xi^\mathfrak{D})_{\mathcal{K}^* \in \mathfrak{M}^*}$:

$$\forall \mathcal{K} \in \mathfrak{M}, \quad \text{rot}^\mathcal{K} \xi^\mathfrak{D} = \frac{1}{m(\mathcal{K})} \sum_{\mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D}_\mathcal{K}} m(\sigma) \xi^\mathfrak{D} \cdot \boldsymbol{\tau},$$

$$\forall \mathcal{K}^* \in \mathfrak{M}^*, \quad \text{rot}^{\mathcal{K}^*} \xi^\mathfrak{D} = \frac{1}{m(\mathcal{K}^*)} \sum_{\mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D}_{\mathcal{K}^*}} m(\sigma^*) \xi^\mathfrak{D} \cdot \boldsymbol{\tau}^*.$$

I.5 Properties of the operators

I.5.1 Discrete Green formulae

Here, we check that the discrete operators verify some discrete duality principles. Les opérateurs discrets introduits sont en dualité. En effet, grâce aux produits scalaires ci-dessus (ref), on peut écrire une formule de Green. C'est cette propriété de dualité qui a donné le nom à la méthode.

Theorem I.5.1 (Discrete Green formulae). Pour tout $\xi^\mathfrak{D}, u^T \in (\mathbb{R}^2)^\mathfrak{D} \times \mathbb{R}^T$, on a

$$(\text{div}^T \xi^\mathfrak{D}, u^T)_T = -(\xi^\mathfrak{D}, \nabla^\mathfrak{D} u^T)_\mathfrak{D},$$

$$(\text{rot}^T \xi^\mathfrak{D}, u^T)_T = (\xi^\mathfrak{D}, \text{rot}^\mathfrak{D} u^T)_\mathfrak{D}.$$

■ **Proof ([DO05, Prop. 2.3]).** Let $\xi^\mathfrak{D}$ be a discrete vector field and $\nabla^\mathfrak{D} u^T$ the discrete gradient of a function u^T , respectively defined by the values $\xi^\mathfrak{D}$ and formula ?? on each of diamond-cells $\mathcal{D} \in \mathfrak{D}$. There holds:

$$(\xi^\mathfrak{D}, \nabla^\mathfrak{D} u^T)_\mathfrak{D} = \sum_{\mathcal{D} \in \mathfrak{D}} m(\mathcal{D}) \xi^\mathfrak{D} \cdot \nabla^\mathfrak{D} u^T = \sum_{\mathcal{D} \in \mathfrak{D}} \frac{1}{2} \xi^\mathfrak{D} \cdot \left[(u_\mathcal{L} - u_\mathcal{K}) m(\sigma) \boldsymbol{\nu} + (u_{\mathcal{L}^*} - u_{\mathcal{K}^*}) m(\sigma^*) \boldsymbol{\nu}^* \right]. \quad (\text{I.5.1})$$

$$(\xi^{\mathfrak{D}}, \nabla^{\mathfrak{D}} u^T)_{\mathfrak{D}} = -\frac{1}{2} \sum_{\kappa \in \mathfrak{M}} u_{\kappa} \left(\sum_{\mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D}_{\kappa}} m(\sigma) \xi^{\mathfrak{D}} \cdot \nu \right) - \frac{1}{2} \sum_{\kappa^* \in \mathfrak{M}^*} u_{\kappa^*} \left(\sum_{\mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D}_{\kappa^*}} m(\sigma^*) \xi^{\mathfrak{D}} \cdot \nu^* \right). \quad (\text{I.5.2})$$

□



Figure I.16

I.5.2 Compositions of the discrete operators [DDO07, Sec. 4.2]

The aim of this section is to **verify** a discrete analogue of the following continuous identities: $\text{div} \text{rot} = 0$, $\text{rot} \nabla = 0$ and $\text{rot} \text{rot} = -\text{div} \nabla$.

Proposition I.5.2. *Given any $u^T \in \mathbb{R}^T$, there holds*

$$\text{div}^T(\text{rot}^{\mathfrak{D}} u^T) = 0 \quad (\text{I.5.3})$$

$$\text{rot}^T(\nabla^{\mathfrak{D}} u^T) = 0. \quad (\text{I.5.4})$$

Proposition I.5.3. *The following equality holds*

$$\text{rot}^T \text{rot}^{\mathfrak{D}} u^T = -\text{div}^T \nabla^{\mathfrak{D}} u^T.$$

I.5.3 Hodge's decomposition

[DDO07, Sec. 4.3] In the continuous case, the Hodge decomposition for non simply connected domains reads:

$$(\mathbb{L}^2)^2 = \nabla V \oplus^{\perp} \text{rot} W, \quad (\text{I.5.5})$$

with $V = \{\phi \in H^1 \mid \int_{\Omega} \phi = 0\}$ and $W = \{\psi \in H^1 \mid \cdot\}$. To prove an analogous property in the discrete case, we rely on the following result:

Lemma I.5.4 (Euler's formula). *For a non simply connected ...*

$$\chi = \dots$$

We may now state the following discrete Hodge decomposition:

Theorem I.5.5. *Let $\xi^{\mathfrak{D}} \in (\mathbb{R}^2)^{\mathfrak{D}}$ be a discrete vector field defined by its values on the diamond-cells $\mathcal{D}$. There exist unique $\phi^T \in \mathbb{R}^T, \psi^T \in \mathbb{R}^T$ such that:*

$$\xi^T = \nabla^{\mathfrak{D}} \phi^T + \text{rot}^{\mathfrak{D}} \psi^T, \quad (\text{I.5.6})$$

$$\sum_{\kappa \in \mathfrak{M}} \alpha_{\kappa} \phi_{\kappa} = \sum_{\kappa^* \in \mathfrak{M}^*} \alpha_{\kappa^*} \phi_{\kappa^*} = 0. \quad (\text{I.5.7})$$

Moreover, the decomposition I.5.6 is orthogonal.

■ **Proof.**

□

Remark I.5.6. Formulae I.5.7 are discrete analogues (respectively stated on the primal mesh and on the dual mesh) of the condition $\int_{\Omega} \phi = 0$ that appears in the definition of the space V in I.5.5.

[GMD25, Sec. 3.2.5] Nous allons aussi définir la divergence et la divergence perp. Pour cela, nous allons utiliser la dualité grâce à la formule d'intégration par parties

$$\left(\operatorname{div}^T \xi^{\mathfrak{D}}, u^T \right)_T = - \left(\xi^{\mathfrak{D}}, \nabla^{\mathfrak{D}} u^T \right)_{\mathfrak{D}}.$$

L'objectif est de résoudre

$$\xi^{\mathfrak{D}} = \nabla^{\mathfrak{D}} \varphi^T + \mathbf{rot}^{\mathfrak{D}} \psi^T + r^{\mathfrak{D}}. \quad (\text{I.5.8})$$

En prenant la divergence, on obtient

$$\operatorname{div}^T \xi^{\mathfrak{D}} = \Delta \varphi^T.$$

On ne va jamais calculer explicitement la divergence et la divergence perp, mais on sera amené à résoudre l'équation

$$\forall v^T \in \mathbb{R}^T, \quad \left(\xi^{\mathfrak{D}}, \nabla^{\mathfrak{D}} v^T \right)_{\mathfrak{D}} = \left(\nabla^{\mathfrak{D}} \varphi^T, \nabla^{\mathfrak{D}} v^T \right)_{\mathfrak{D}}.$$

Pour obtenir l'équation en ψ^T , on applique rot^T à l'équation I.5.8:

...

Remark I.5.7. La matrice $\mathbf{R}_{\mathfrak{D}}^T \mathbf{M}_{\mathfrak{D}}$ est symétrique positive mais pas définie. Néanmoins, cela n'est pas gênant car la méthode du gradient conjugué converge à un vecteur du noyau de $\mathbf{G}_{\mathfrak{D}}$ près. Or le noyau de $\mathbf{G}_{\mathfrak{D}}$ contient l'homologie ainsi que des modes hyper-oscillants. De plus, la composante dans le noyau de $\mathbf{G}$ est préservée dans toutes les étapes du gradient conjugué et en prenant comme solution initiale un vecteur nul, cela nous assure de n'obtenir aucun de ces modes tout en convergeant.

I.6 Regularity of meshes

- On note $\sigma_{\mathcal{K}} \stackrel{\text{déf.}}{=} B(x_{\mathcal{K}}, \rho_{\mathcal{K}})$ et $\sigma_{\mathcal{K}^*} \stackrel{\text{déf.}}{=} B(x_{\mathcal{K}^*}, \rho_{\mathcal{K}^*})$ and $\rho_{\mathcal{K}}$ and $\rho_{\mathcal{K}^*}$ are positive numbers associated to the mesh $\mathcal{T}$ and such that

$$\sigma_{\mathcal{K}} \subset \bar{\mathcal{K}}, \quad \sigma_{\mathcal{K}^*} \subset \partial \mathcal{K}^*.$$

Finally, introduce numbers $\rho_{\mathcal{K}}$ and $\rho_{\mathcal{K}^*}$ for any $\mathcal{K} \in \mathfrak{M}$ (resp. $\mathcal{K}^* \in \mathfrak{M}^*$) such that

$$B_{\mathcal{K}} \stackrel{\text{déf.}}{=} B(x_{\mathcal{K}}, \rho_{\mathcal{K}}) \subset \Omega, \quad B_{\mathcal{K}^*} \stackrel{\text{déf.}}{=} B(x_{\mathcal{K}^*}, \rho_{\mathcal{K}^*}) \subset \Omega.$$

These balls are only introduced in order to prove the convergence of the scheme.

We note $\operatorname{size}(\mathcal{T})$ the maximum of the diameters of the diamond cells in $\mathfrak{D}$:

$$\operatorname{size}(\mathcal{T}) \stackrel{\text{déf.}}{=} \max_{\mathcal{D} \in \mathfrak{D}} d_{\mathcal{D}}.$$

The following bounds follow:

$$m(\sigma) \leq \operatorname{size}(\mathcal{T}), \quad \forall \sigma \in \mathfrak{E}; \quad m(\sigma^*) \leq \operatorname{size}(\mathcal{T}), \quad \forall \sigma^* \in \mathfrak{E}^*;$$

$$m(\mathcal{K}) \leq \pi \operatorname{size}(\mathcal{T})^2, \quad \forall \mathcal{K} \in \mathfrak{M}; \quad m(\mathcal{K}^*) \leq \pi \operatorname{size}(\mathcal{T})^2, \quad \forall \mathcal{K}^* \in \mathfrak{M}^*; \quad m(\mathcal{D}) \leq \frac{1}{2} \operatorname{size}(\mathcal{T})^2, \quad \forall \mathcal{D} \in \mathfrak{D}.$$

We introduce now a positive number that quantifies the regularity of a given mesh and is useful to perform the convergence analysis of the finite volume schemes. We first define $\alpha_{\mathcal{T}}$ to be the unique real number in $]0; \frac{\pi}{2}]$ such that

$$\sin \alpha_{\mathcal{T}} \stackrel{\text{déf.}}{=} \min_{\mathcal{D} \in \mathfrak{D}} |\sin \alpha_{\mathcal{D}}|,$$

that is the minimal angle between the diagonals of the diamond cells in the mesh.

Let us introduce the number

$$\begin{aligned} \mathcal{N}_{\mathcal{T}} \stackrel{\text{def.}}{=} & \sup_{x \in \Omega} \# \left\{ \kappa^* \mid x \in \widehat{\kappa^* \cup B_{\kappa^*}} \right\} + \sup_{x \in \Omega} \# \left\{ \kappa \mid x \in \widehat{\kappa \cup B_{\kappa}} \right\} \\ & + \sup_{x \in \Omega} \# \left\{ \mathcal{D} \mid x \in \widehat{\mathcal{D} \cup B_{\kappa}}, \mathcal{D} \in \mathfrak{D}_{\kappa}, \kappa \in \mathfrak{M} \right\} \\ & + \sup_{x \in \Omega} \# \left\{ \mathcal{D} \mid x \in \widehat{\mathcal{D} \cup B_{\kappa^*}}, \mathcal{D} \in \mathfrak{D}_{\kappa^*}, \kappa^* \in \mathfrak{M}^* \right\}, \end{aligned}$$

where $\widehat{E}$ denotes the convex hull of any set $E \subset \mathbb{R}^2$. In the usual case where the primal control volumes, the dual control volumes and the diamond cells are **convex** and where $B_{\kappa} \subset \kappa$, $B_{\kappa^*} \subset \kappa^*$ then $\mathcal{N}_{\mathcal{T}}$ can be bounded by a function of the maximal number of edges per primal and dual control volumes. Since we do not impose any convexity assumption on the meshes, we need to control this number $\mathcal{N}_{\mathcal{T}}$ in the convergence analysis. We can now introduce

$$\begin{aligned} \text{reg } \mathcal{T} \stackrel{\text{def.}}{=} & \max \left\{ \frac{1}{\alpha_{\mathcal{T}}}, \mathcal{N}_{\mathcal{T}}, \max_{\mathcal{D} \in \mathfrak{D}} \frac{d_{\mathcal{D}}}{\sqrt{m(\mathcal{D})}}, \max_{\kappa \in \mathfrak{M}} \frac{d_{\kappa}}{\sqrt{m(\kappa)}}, \max_{\kappa^* \in \mathfrak{M}^*} \frac{d_{\kappa^*}}{\sqrt{m(\kappa^*)}}, \right. \\ & \left. \max_{\kappa \in \mathfrak{M}} \left(\frac{d_{\kappa}}{\rho_{\kappa}} + \frac{\rho_{\kappa}}{d_{\kappa}} \right), \max_{\kappa^* \in \mathfrak{M}^*} \left(\frac{d_{\kappa^*}}{\rho_{\kappa^*}} + \frac{\rho_{\kappa^*}}{d_{\kappa^*}} \right), \max_{\substack{\kappa \in \mathfrak{M} \\ \mathcal{D} \in \mathfrak{D}_{\kappa}}} \frac{d_{\kappa}}{d_{\mathcal{D}}}, \max_{\substack{\kappa^* \in \mathfrak{M}^* \\ \mathcal{D} \in \mathfrak{D}_{\kappa^*}}} \frac{d_{\kappa^*}}{d_{\mathcal{D}}} \right\}. \end{aligned}$$

Let us point out that $\text{reg}(\mathcal{T})$ essentially measures:

- how flat the diamond cells are;
- how large is the difference between the size of a primal control volume (resp. a dual control volume) and the size of a diamond cell as soon as they intersect.

Convention. In any estimate given, the dependence of the constants in $\text{reg}(\mathcal{T})$ is implicitly assumed to be non-decreasing. Furthermore, the dependence of the constants on the domain is often omitted.

Remark I.6.1. For conformal finite volume meshes, it is assumed that $\alpha_{\mathcal{D}} = \frac{\pi}{2}$ for any diamond $\mathcal{D} \in \mathfrak{D}$, so that $\alpha_{\mathcal{T}} = \frac{\pi}{2}$. In our case, not only this orthogonality condition is relaxed but also $\mathfrak{M}$ can present atypical edges, non convex dual control volumes and non convex diamond cells [figure](#).

Remark I.6.2. The boundedness of $\text{reg}(\mathcal{T})$ imposes only local restriction on the mesh. It is easy to construct a family of locally refined mesh such that $\text{reg}(\mathcal{T})$ is bounded independently on the level of the refinement.

I.6.1 Régularité du maillage [Kre10, p. 36]

Soit $\text{size}(\mathcal{T})$ le maximum des diamètres des diamants. Pour mesurer l'applatissage des diamants, on note $\alpha_{\mathcal{T}}$ l'unique réel dans $]0; \frac{\pi}{2}]$ tel que $\sin(\alpha_{\mathcal{T}}) \stackrel{\text{def.}}{=} \min_{\mathcal{D} \in \mathfrak{D}} |\sin(\alpha_{\mathcal{D}})|$ si la maillage dual est **direct** et $\sin(\alpha_{\mathcal{T}}) \stackrel{\text{def.}}{=} \min_{\mathcal{D} \in \mathfrak{D}} (|\sin(\alpha_{\kappa})|, |\sin(\alpha_{\kappa^*})|)$ si le maillage dual est **barycentrique**. On introduit un nombre positif $\text{reg}(\mathcal{T})$ qui mesure la régularité d'un maillage donné et qui est utilisé pour réaliser la convergence des schémas:

$$\begin{aligned} \mathcal{N}_{\mathcal{T}} = & \sup_{x \in \Omega} \# \left\{ \mathcal{D} \mid x \in \widehat{\mathcal{D} \cup \kappa}, \mathcal{D} \in \mathfrak{D}_{\kappa}, \kappa \in \mathfrak{M} \right\} + \sup_{x \in \Omega} \# \left\{ \mathcal{D} \mid x \in \widehat{\mathcal{D} \cup \kappa^*}, \mathcal{D} \in \mathfrak{D}_{\kappa^*}, \kappa^* \in \mathfrak{M}^* \right\}, \\ \text{reg}(\mathcal{T}) = & \max \left\{ \frac{1}{\sin(\alpha_{\mathcal{T}})}, \mathcal{N}, \mathcal{N}^*, \mathcal{N}_{\mathcal{T}}, \max_{\mathcal{D} \in \mathfrak{D}} \max_{\substack{\mathcal{Q} \in \mathfrak{Q}_{\mathcal{D}} \\ \sigma \in \partial \mathcal{Q}}} \frac{d_{\mathcal{D}}}{m(\sigma)}, \max_{\kappa \in \mathfrak{M}} \frac{d_{\kappa}}{\sqrt{m(\kappa)}}, \max_{\kappa^* \in \mathfrak{M}^*} \frac{d_{\kappa^*}}{\sqrt{m(\kappa^*)}}, \right. \\ & \left. \max_{\kappa \in \mathfrak{M}} \max_{\mathcal{D} \in \mathfrak{D}_{\kappa}} \frac{d_{\kappa}}{d_{\mathcal{D}}}, \max_{\kappa^* \in \mathfrak{M}^*} \max_{\mathcal{D} \in \mathfrak{D}_{\kappa^*}} \frac{d_{\kappa^*}}{d_{\mathcal{D}}} \right\}, \end{aligned}$$

où $\widehat{P}$ désigne l'enveloppe convexe d'un ensemble P , $\mathcal{N}$, $\mathcal{N}^*$ sont le nombre maximum d'arêtes d'une maille primale et le nombre maximum d'arêtes incidentes d'un sommet du maillage primal. Ce nombre $\text{reg}(\mathcal{T})$ mesure essentiellement l'applatissage des diamants et le rapport entre la taille des mailles primales (resp. duales) et la taille des diamants et

il doit être uniformément borné quand $\text{size}(\mathcal{T}) \rightarrow 0$ pour avoir la convergence. Par exemple, ce nombre $\text{reg}(\mathcal{T})$ implique les résultats géométriques suivants: il existe une constante $C > 0$, dépendant uniquement de $\text{reg}(\mathcal{T})$ telle que

$$\frac{d_{\mathcal{D}}}{\sqrt{m(\mathcal{D})}} \leq C, \forall \mathcal{D} \in \mathfrak{D}, \quad \frac{d_{\mathcal{Q}}}{\sqrt{m(\mathcal{Q})}} \leq C, \forall \mathcal{Q} \in \mathfrak{Q} \quad \text{et} \quad d_{\mathcal{D}} \leq C \min(m(\sigma), m(\sigma^*)), \forall \mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D}.$$

Hypothèses. On suppose que toute maille primale $\kappa \in \mathfrak{M}$ est étoilée par rapport à x_{κ} .

On suppose que toute maille duale $\kappa^* \in \mathfrak{M}^*$ est étoilée par rapport à x_{κ^*} .

Remark I.6.3. On a que toute maille diamant $\mathcal{D}$ est étoilée par rapport à $x_{\mathcal{D}}$.

Remark I.6.4. Grâce aux hypothèses ci-dessus, on a également:

$$\text{diam}(\hat{\kappa}) \leq 2 d_{\kappa}, \forall \kappa \in \mathfrak{M} \quad \text{et} \quad \text{diam}(\hat{\kappa}^*) \leq 2 d_{\kappa^*}, \forall \kappa^* \in \mathfrak{M}^*,$$

de plus la remarque I.6.3 implique:

$$\text{diam}(\hat{\mathcal{D}}) \leq 2 d_{\mathcal{D}}, \forall \mathcal{D} \in \mathfrak{D}.$$

I.7 Estimates for the geometric errors [DE13, Sec. 4.2]

We consider a piecewise linear approximation of a smooth surface $\mathcal{M} \in \mathcal{C}^2$ and its geometry. We collect estimates for the geometric error which is produced by approximating the smooth surface $\mathcal{M}$ by a discrete surface $\mathcal{M}_h$.

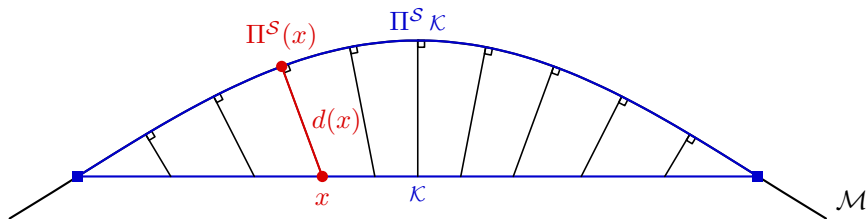


Figure I.17. A curved simplex $\Pi^{\mathcal{S}} \kappa$ is parametrized over a planar one. Orthogonal projection onto the smooth surface $\mathcal{M}$

Lemma I.7.1 (Projection [DE13, Lemma 4.1]). Assume $\mathcal{M}$ and $\mathcal{M}_h$ are as above. The map $\Pi^{\mathcal{S}}: \mathcal{M}_h \rightarrow \mathcal{M}$ is bijective. For the oriented distance function to $\mathcal{M}$ we have the estimate

$$\|d\|_{L^\infty(\mathcal{M}_h)} \leq c h^2.$$

The quotient, δ_h , between the smooth and discrete surface measures dA and dA_h , defined by $\delta_h dA_h = dA$, satisfies

$$\|1 - \delta_h\|_{L^\infty(\mathcal{M}_h)} \leq c h^2.$$

Let P and P_h be the projections onto the tangent planes, $P_{ij} = \delta_{ij} - \nu_i \nu_j$, $P_{h,ij} = \delta_{ij} - \nu_{h,i} \nu_{h,j}$, and let

$$R_h \stackrel{\text{d\'ef.}}{=} \frac{1}{\delta_h} P(I - d\mathcal{H}) P_h (I - d\mathcal{H}),$$

$\mathcal{H}_{ij} = d_{x_i x_j} = \nu_{i x_j}$. Then

$$\|(I - R_h)P\|_{L^\infty(\mathcal{M}_h)} \leq c h^2.$$

Lemma I.7.2.

$$\|d\|_{\mathfrak{M} + \mathfrak{M}^*} \leq c \text{size}(\mathcal{T})^2.$$

Lemma I.7.3. *À plat, $m(\text{arête}) \sim h$ et $m(\text{triangle}) \sim h^2$. Sur la surface aussi.*

For any function η defined on the discrete surface $\mathcal{M}_h$ we define an extension or lift onto $\mathcal{M}$ by

$$\eta^\ell(a) = \eta(x(a)), \quad a \in \mathcal{M}, \quad (\text{I.7.1})$$

where, by our assumptions, $x(a)$ is defined as the unique solution of

$$x = a + d(x) \nu(a).$$

Furthermore, we understand by $\eta^\ell(x)$ the constant extension from $\mathcal{M}$ in the normal direction $\nu(a(x))$.

In order to compare the norms between functions on $\mathcal{M}_h$ and their lift I.7.1 to $\mathcal{M}$ we need the following lemma.

Lemma I.7.4. *Let $\eta: \mathcal{M}_h \rightarrow \mathbb{R}$ with the lift $\eta^\ell: \mathcal{M} \rightarrow \mathbb{R}$. Then, for the plane $T \subset \mathcal{M}_h$, and smooth curved triangles $\tilde{T} \subset \mathcal{M}$, the following estimates hold if the norm exist. There is a constant $c > 0$ independant of h such that*

$$\begin{aligned} \frac{1}{c} \|\eta\|_{L^2(T)} &\leq \|\eta^\ell\|_{L^2(\tilde{T})} \leq c \|\eta\|_{L^2(T)}, \\ \frac{1}{c} \|\nabla_{\mathcal{M}_h} \eta\|_{L^2(T)} &\leq \|\nabla_{\mathcal{M}} \eta^\ell\|_{L^2(\tilde{T})} \leq c \|\nabla_{\mathcal{M}_h} \eta\|_{L^2(T)}, \\ \|\nabla_{\mathcal{M}_h}^2 \eta\|_{L^2(T)} &\leq c \|\nabla_{\mathcal{M}}^2 \eta^\ell\|_{L^2(\tilde{T})} + c h \|\nabla_{\mathcal{M}} \eta^\ell\|_{L^2(\tilde{T})}. \end{aligned}$$

I.7.1 Discrete charts [DE13, Sec. 4.3]

We add some information on a discrete differential geometric approach which is consistent with the introduction of parametrized surfaces. For a complete exposition see Nédélec (1976).

Let $\mathcal{M}$ be given as in Section 2.1 by local charts $(X_i)_{i \in I}$, $X_i \in \mathcal{C}^k(V_i, \mathbb{R}^{n+1})$. We assume that already

$$\mathcal{M} = \bigcup_{i \in I} X_i(V_{ih}),$$

with $V_{ih} \subset V_i$ and triangulated domains $V_{ih} \subset \mathbb{R}^n$. We interpolate the smooth parametrization $X_{ih} = I_h X_i$, where I_h is the interpolation from Lemma 4.3 on the polygonal domain V_{ih} . We assume that if $U_{ij} = X_{ih}(V_{ih}) \cap X_{jh}(V_{jh}) \neq \emptyset$, then the map $X_{ih}^{-1} \circ X_{jh}: X_{jh}^{-1}(U_{ij}) \rightarrow X_{ih}^{-1}(U_{ij})$ is bijective, continuous and piecewise linear. The discrete surface is defined by

$$\mathcal{M}_h = \bigcup_{i \in I} X_{ih}(V_{ih}).$$

The advantage of this way to discretize the surface $\mathcal{M}$ is that it allows the approximation of immersions.

Chapter II

Inégalité de POINCARÉ DDFV sur une surface fermée sans bord de $\mathbb{R}^3$

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II.1 The Isoperimetric Problem and the Cheeger Constant [Ben16, Sec. 3]

Remark II.1.1. The following results are given in dimension $n \geq 1$ but will then be applied for $n = 2$ which our setting.

Let $\mathcal{M}$ be an n -dimensional, Riemannian manifold with $\text{vol}_n(\mathcal{M}) < \infty$. For a positive n -Hausdorff measure subset $A \subseteq \mathcal{M}$ having $(n-1)$ -Hausdorff measurable boundary ∂A , the **isoperimetric ratio** of A is given by

$$h^*(A) \stackrel{\text{def.}}{=} \frac{\text{vol}_{n-1}(\partial A)}{\text{vol}_n(A)}. \quad (\text{II.1.1})$$

The isoperimetric ratio is well-defined when A is a smooth submanifold with boundary of $\mathcal{M}$ or, more generally, when A is an integral current. We also define the **isoperimetric constant** of a subset $A \subseteq \mathcal{M}$ to be

$$\bar{h}(A) = \inf_{D \subseteq A} \frac{\text{vol}_{n-1}(\partial D)}{\text{vol}_n(D)} = \inf_{D \subseteq A} h^*(D), \quad (\text{II.1.2})$$

where D ranges over all subsets of A which have smooth boundary.

For $t \in]0; \text{vol}_n(\mathcal{M})[$, the isoperimetric problem of volume t studies the problem of finding the subset A of $\mathcal{M}$ with $\text{vol}_n(A) = t$ and

$$\text{vol}_{n-1}(\partial A) = \inf \{ \text{vol}_{n-1}(\partial B) \mid \text{vol}_n(B) = t \}.$$

We call such an A an **isoperimetric minimizer**. For an in-depth exposition on the isoperimetric problem for Riemannian manifolds, see [Ros05].

The Cheeger constant is related to the isoperimetric problem in that we wish to minimize the isoperimetric ratio among all isoperimetric problems corresponding to volumes $t \in]0; \text{vol}_n(\mathcal{M})/2]$. We will call a set D in $\mathcal{M}$ with $h^*(D)$ well-defined and $\text{vol}_n(D) \leq \text{vol}_n(\mathcal{M})/2$ a **Cheeger candidate**.

As described by [Ros05], one obtains the **fundamental existence and regularity theorem** for isoperimetric minimizers of Riemannian manifolds.

Theorem II.1.2 (Fundamental Existence and Regularity). When $\mathcal{M}$ is a **smooth, compact**, possibly with boundary, and $t \in]0; \text{vol}_n(\mathcal{M})[$, there exists a compact region $A \subset \mathcal{M}$ such that ∂A minimizes the $(n-1)$ -volume among all regions of n -volume t . Further, for any such minimizer, ∂A is a smooth embedded hypersurface of constant mean curvature, except for a singular set of Hausdorff dimension at most $n-8$. Finally, if $\partial\mathcal{M} \cap \partial A \neq \emptyset$, then $\partial\mathcal{M}$ and ∂A meet orthogonally.

As suggested by II.1.2 and smooth approximation techniques, it is equivalent to minimize the isoperimetric ratio over smooth submanifolds with boundary or the more general collection of rectifiable currents to find the Cheeger constant. In the same way that we have defined isoperimetric minimizers, we can define minimizers for the Cheeger constant, which we refer to as Cheeger minimizers for brevity. Specifically, a **Cheeger minimizer** is a subset A of $\mathcal{M}$ such that $\text{vol}_n(A) \leq \text{vol}_n(\mathcal{M})/2$ and $h^*(A) = h(\mathcal{M})$. When $\mathcal{M}$ is compact, Buser showed that there exist volume(s) $t \in]0; \text{vol}_n(\mathcal{M})/2]$ so that Cheeger minimizers coincide with solutions to the isoperimetric problem of n -volume t in $\mathcal{M}$ [Bus82, Remark 3.3, Lemma 3.4]. Therefore, the existence and regularity of a Cheeger minimizer follows from II.1.2. Specifically, Buser proved:

Lemma II.1.3 (Buser, [Bus82, Lemma 3.4]). Let $\mathcal{M}$ be an n -dimensional, compact Riemannian manifold, possibly with boundary. Then there exists $t^* \in]0; \text{vol}_n(\mathcal{M})/2]$ and a sequence of submanifolds $A_k \subset \mathcal{M}$ with smooth boundary such that $\text{vol}_n(A_k) = t^*$ for all k and

$$h(\mathcal{M}) = \lim_{k \rightarrow \infty} \frac{\text{vol}_{n-1}(\partial A_k)}{\text{vol}_n(A_k)}.$$

Combining II.1.2 and II.1.3, we can immediately conclude that **Cheeger minimizers exists for all compact manifolds**.

Interprétation. « On ne peut pas trouver d'ensembles de périmètres qui tendent vers 0 et de volumes non nuls ».

II.2 Graphs

II.2.1 Laplacian of a graph [Chu, Sec. 1.2]

In a graph G , let d_v denote the degree of the vertex v . We first define the Laplacian for graphs without loops and multiple edges. To begin, we consider the matrix L , defined as follows:

$$L(u, v) = \begin{cases} d_v & \text{if } u = v \\ -1 & \text{if } u \text{ and } v \text{ are adjacent,} \\ 0 & \text{otherwise.} \end{cases}$$

Let T denote the diagonal matrix with the (v, v) -th entry having value d_v . The *Laplacian* of G is defined to be the matrix

$$\mathcal{L}(u, v) = \begin{cases} 1 & \text{if } u = v \text{ and } d_v \neq 0 \\ -\frac{1}{\sqrt{d_u d_v}} & \text{if } u \text{ and } v \text{ are adjacent,} \\ 0 & \text{otherwise.} \end{cases}$$

We can write

$$\mathcal{L} = T^{-1/2} L T^{-1/2}$$

with the convention $T^{-1}(v, v) = 0$ for $d_v = 0$. We say v is an isolated vertex if $d_v = 0$. A graph is said to be nontrivial if it contains at least one edge. The Laplacian $\mathcal{L}$ can be viewed as an operator on the space of functions $g: V(G) \rightarrow \mathbb{R}$ which satisfies

$$\mathcal{L}g(u) = \frac{1}{\sqrt{d_u}} \sum_{v \sim u} \left(\frac{g(u)}{\sqrt{d_u}} - \frac{g(v)}{\sqrt{d_v}} \right).$$

When G is k -regular, it is easy to see that

$$\mathcal{L} = I - \frac{1}{k} A,$$

where A is the adjacency matrix of G and I is an identity matrix. All matrices here are $n \times n$ where n is the number of vertices of G .

For a general graph, we have

$$\mathcal{L} = T^{-1/2} L T^{-1/2} = \mathbf{I} - T^{-1/2} A T^{-1/2}.$$

(lien avec la théorie de l'homologie)

II.2.2 Eigenvalues of a graph [Chu, Sec. 1.2]

Since $\mathcal{L}$ is symmetric, its eigenvalues are all real and non-negative. We can use the variational characterizations of those eigenvalues in terms of the Rayleigh quotient of $\mathcal{L}$. Let g denote an arbitrary function which assigns to each vertex v of G a real value $g(v)$. We can view g as a column vector. Then

$$\begin{aligned} \frac{\langle g, \mathcal{L}g \rangle}{\langle g, g \rangle} &= \frac{\langle g, T^{-1/2} L T^{-1/2} g \rangle}{\langle g, g \rangle} \\ &= \frac{\langle f, Lf \rangle}{\langle T^{1/2} f, T^{1/2} f \rangle} \\ &= \frac{\sum_{u \sim v} (f(u) - f(v))^2}{\sum_v f(v)^2 d_v} \end{aligned} \quad (\text{II.2.1})$$

where $g = T^{1/2} f$ and $\sum_{u \sim v}$ denotes the sum over all unordered pairs $\{u, v\}$ for which u and v are adjacent. Here $\langle f, g \rangle = \sum_x f(x) g(x)$ denotes the standard inner product in $\mathbb{R}^n$. The sum $\sum_{u \sim v} (f(u) - f(v))^2$ is sometimes called the *Dirichlet sum* of G and the ratio on the left-hand side of II.2.1 is often called the *Rayleigh quotient*.

From equation II.2.1, we see that all eigenvalues are non-negative. In fact, we can easily deduce from equation II.2.1 that 0 is an eigenvalue of $\mathcal{L}$. We denote the eigenvalues of $\mathcal{L}$ by $0 = \lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_{n-1}$. The set of the λ_i 's is called the *spectrum* of $\mathcal{L}$. Let $\mathbf{1}$ denote the constant function which assumes the value 1 on each vertex. Then $T^{1/2} \mathbf{1}$ is an eigenfunction of $\mathcal{L}$ with eigenvalue 0. Furthermore,

$$\lambda_G = \lambda_1 = \inf_{f \perp T \mathbf{1}} \frac{\sum_{u \sim v} (f(u) - f(v))^2}{\sum_v f(v)^2 d_v}. \quad (\text{II.2.2})$$

(f ranges over functions defined on the vertices of G satisfying $\sum_v f(v) d_v = 0$). The corresponding eigenfunction is $g = T^{1/2} f$ as in II.2.1. The spectral gap λ_G is the least nonzero eigenvalue of $\mathcal{L}$.

The above formulation for λ_G corresponds in a natural way to the eigenvalues of the Laplace–Beltrami operator for Riemannian manifolds:

$$\lambda_M = \inf_{\int_M f = 0} \frac{\int_M |\nabla f|^2}{\int_M |f|^2}.$$

We remark that the corresponding measure here for each edge is 1. The measure for each vertex is the degree of the vertex.

We note that II.2.2 also has the formulation

$$\lambda_G = \lambda_1 = \inf_f \frac{\sum_{u \sim v} (f(u) - f(v))^2}{\sum_v (f(v) - \bar{f})^2 d_v}, \quad \text{where } \bar{f} = \frac{\sum_v f(v) d_v}{\text{vol } G}, \quad (\text{II.2.3})$$

and $\text{vol } G$ denotes the volume of the graph G , given by $\text{vol } G = \sum_v d_v$. More generally, for a set S , the volume of S , denoted by $\text{vol}(S)$ is defined to be $\text{vol}(S) \stackrel{\text{def}}{=} \sum_{v \in S} d_v$.

II.2.3 Cheeger inequality on graphs [Ji+10, Sec. 2]

For a set S , the *Cheeger ratio* of S is defined to be

$$h_S \stackrel{\text{def}}{=} \frac{|\partial S|}{\min\{\text{vol}(S), \text{vol}(G) - \text{vol}(S)\}},$$

where the boundary of S is denoted $\partial S \stackrel{\text{def.}}{=} \{\{u, v\} \in E \mid u \in S \text{ and } v \notin S\}$. Let $\bar{S}$ denote the complement of S . Then $h_S = h_{\bar{S}}$. The *Cheeger constant* of G is denoted by

$$h_G \stackrel{\text{def.}}{=} \min_S h_S. \quad (\text{II.2.4})$$

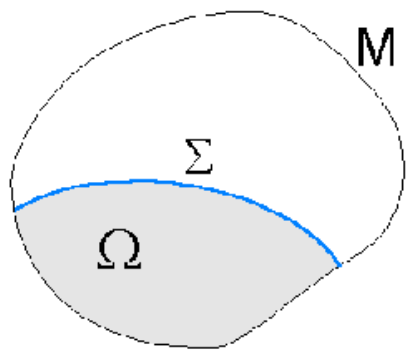


Figure II.1

Theorem II.2.1 (Cheeger inequality [Ji+10, Theo. 1]). In a graph G , the Cheeger constant h_G and the spectral gap λ_G are related as follows:

$$2h_G \geq \lambda_G \geq \frac{\alpha_G^2}{2} \geq \frac{h_G^2}{2}$$

where α_G is the minimum Cheeger ratio of subsets S_i consisting of vertices *with the largest i values in the eigenvector associated with λ_G , over all i .*

■ **Proof.** For a proof of the Cheeger inequality using eigenvectors, see [Ji+10, Sec. 3]. □

II.3 Main theorem

II.3.1 Notations & Énoncé

Lemma II.3.1. *Il existe $\underline{C}^T, \overline{C}^T, \underline{C}^{\mathcal{D}}$ et $\overline{C}^{\mathcal{D}}$ indépendantes de h et dépendantes de $\text{reg } \mathcal{T}$ (refaire la démo de [DE13] pour bien comprendre de quoi dépendent ces constantes) telles que*

$$\begin{aligned} \forall \kappa \in \mathfrak{M}, \quad 2 \underline{C}^T \mu_\kappa &\leq m(\kappa) \leq 2 \overline{C}^T \mu_\kappa, \\ \forall \kappa^* \in \mathfrak{M}^*, \quad 2 \underline{C}^T \mu_{\kappa^*} &\leq m(\kappa^*) \leq 2 \overline{C}^T \mu_{\kappa^*}, \\ \forall \mathcal{D} \in \mathfrak{D}, \quad \underline{C}^{\mathcal{D}} \mu_{\mathcal{D}} &\leq m(\mathcal{D}) \leq \overline{C}^{\mathcal{D}} \mu_{\mathcal{D}}. \end{aligned}$$

Proposition II.3.2.

$$\|\cdot\|_{2,\mathfrak{D}} \sim \|\cdot\|_{L^2(\mathcal{M})}$$

Proposition II.3.3.

$$\underline{C}^T \leq \|\Pi_{\mathcal{T}}^S\|_{L^2} \leq \overline{C}^T, \quad \underline{C}^{\mathcal{D}} \leq \|\Pi_{\mathfrak{D}}^S\|_{L^2} \leq \overline{C}^{\mathcal{D}}. \quad (\text{II.3.1})$$

■ **Proof.**

$$\|\Pi_{\mathcal{T}}^S\|_{L^2} = \sup_{u^T \in \mathbb{R}^T} \frac{\|\Pi_{\mathcal{T}}^S u^T\|_{L^2(\mathcal{M})}}{\|u^T\|_{2,\mathcal{T}}}$$

□

Nous sommes maintenant en mesure d'énoncer le résultat principal de ce chapitre, à savoir une inégalité de Poincaré DDFV sur une surface fermée sans bord de $\mathbb{R}^3$.

Theorem II.3.4 (Main). *Soit $\mathcal{T}$ un maillage DDFV de $\mathcal{M}$. Il existe une constante C , ne dépendant que de la régularité du maillage $\text{reg}(\mathcal{T})$, telle que pour tout $u^T \in \mathbb{R}^T$, on a*

$$\|u^T\|_{2,\mathcal{T}} \leq C \|\nabla^{\mathcal{D}} u^T\|_{2,\mathfrak{D}}. \quad (\text{II.3.2})$$

De manière équivalente ?, on a

$$\|\Pi_{\mathcal{T}}^S u^T\|_{L^2(\mathcal{M})} \leq C \frac{\overline{C}^{\mathcal{D}}}{\underline{C}^T} \|\Pi_{\mathfrak{D}}^S \nabla^{\mathcal{D}} u^T\|_{L^2(\mathcal{M})}.$$

II.3.2 Une inégalité de graphe

Definition II.3.5. Soit $\mathcal{T} = (\mathfrak{M}, \mathfrak{M}^*)$ un maillage DDFV. On lui associe deux graphes:

- le graphe du maillage primal $\mathcal{G}^{\mathfrak{M}} = (\mathfrak{B}, \mathfrak{E}^*)$ où $\mathfrak{B}$ est l'ensemble des sommets de $\mathfrak{M}^*$ et $\mathfrak{E}^*$ est l'ensemble des arêtes de $\mathfrak{M}^*$;
- le graphe du maillage dual $\mathcal{G}^{\mathfrak{M}^*} = (\mathfrak{B}^*, \mathfrak{E})$ où $\mathfrak{B}^*$ est l'ensemble des sommets de $\mathfrak{M}$ et $\mathfrak{E}$ est l'ensemble des arêtes de $\mathfrak{M}$.

Notations. On note $V(\mathcal{G})$ l'ensemble des sommets d'un graphe $\mathcal{G}$ et $E(\mathcal{G})$ l'ensemble de ses arêtes. On pourra également utiliser la notation V et E lorsqu'il n'y a pas d'ambiguïté sur le graphe.

Soit $u^T = (u^{\mathfrak{M}}, u^{\mathfrak{M}^*}) \in \mathbb{R}^T$. On considère la fonction $u^{\mathfrak{M}}$ comme une fonction sur la graphe $\mathcal{G}^{\mathfrak{M}}$ dans le sens où pour $x_\kappa \in \mathfrak{B}$, on note $u_\kappa = u^{\mathfrak{M}}(x_\kappa)$ et on considère la fonction $u^{\mathfrak{M}^*}$ comme une fonction sur la graphe $\mathcal{G}^{\mathfrak{M}^*}$ dans le sens où pour $x_{\kappa^*} \in \mathfrak{B}^*$, on note $u_{\kappa^*} = u^{\mathfrak{M}^*}(x_{\kappa^*})$.

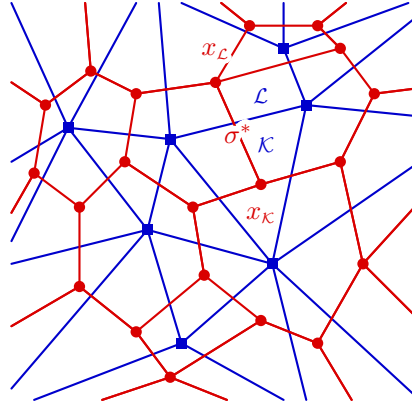


Figure II.2. Maillage primal et graphe associé

Pour tout $u^T \in \mathbb{R}^T$, on pose

$$\mathcal{E}(u^T) \stackrel{\text{d\'ef.}}{=} \sum_{\kappa \sim \mathcal{L}} (u_\kappa - u_{\mathcal{L}})^2 + \sum_{\kappa^* \sim \mathcal{L}^*} (u_{\kappa^*} - u_{\mathcal{L}^*})^2.$$

Pour obtenir Equation (II.3.2), l'objectif va être d'obtenir d'une part $\mathcal{E}(u^T) \gtrsim \|u^T\|_{2,T}^2$ et d'autre part $\mathcal{E}(u^T) \lesssim \|\nabla^{\mathcal{D}} u^T\|_{2,\mathcal{D}}^2$.

Lemma II.3.6 (Une inégalité de graphe). Pour tout $u^T \in \mathbb{R}^T$,

$$\mathcal{E}(u^T) \geq \frac{h_{\mathcal{G}^T}^2}{2} \left(\sum_{\kappa \in \mathfrak{M}} (u_\kappa - c^{\mathfrak{M}})^2 \deg(x_\kappa) + \sum_{\kappa^* \in \mathfrak{M}^*} (u_{\kappa^*} - c^{\mathfrak{M}^*})^2 \deg(x_{\kappa^*}) \right),$$

où

$$c^{\mathfrak{M}} = \frac{\sum_{\kappa \in \mathfrak{M}} u_\kappa \deg(x_\kappa)}{\sum_{\kappa \in \mathfrak{M}} \deg(x_\kappa)}, \quad c^{\mathfrak{M}^*} = \frac{\sum_{\kappa^* \in \mathfrak{M}^*} u_{\kappa^*} \deg(x_{\kappa^*})}{\sum_{\kappa^* \in \mathfrak{M}^*} \deg(x_{\kappa^*})}, \quad \text{et} \quad h_{\mathcal{G}^T} = \min\{h_{\mathcal{G}^{\mathfrak{M}}}, h_{\mathcal{G}^{\mathfrak{M}^*}}\}.$$

La constante $h_{\mathcal{G}^{\mathfrak{M}}}$ (resp. $h_{\mathcal{G}^{\mathfrak{M}^*}}$) est la constante de Cheeger du graphe $\mathcal{G}^{\mathfrak{M}}$ (resp. $\mathcal{G}^{\mathfrak{M}^*}$) définie par Equation (II.2.4).

■ **Proof.** Soit $u^T \in \mathbb{R}^T$. On considère cette fonction comme une fonction sur les graphes $\mathcal{G}^{\mathfrak{M}}$ et $\mathcal{G}^{\mathfrak{M}^*}$ comme expliqué dans la Definition II.3.5. Il suffit alors d'appliquer le Theorem II.2.1, établissant l'inégalité de Cheeger, d'une part au graphe $\mathcal{G}^{\mathfrak{M}}$:

$$\sum_{\kappa \sim \mathcal{L}} (u_\kappa - u_{\mathcal{L}})^2 \geq \frac{h_{\mathcal{G}^{\mathfrak{M}}}^2}{2} \sum_{\kappa \in \mathfrak{M}} (u_\kappa - c^{\mathfrak{M}})^2 \deg(x_\kappa), \quad (\text{II.3.3})$$

et d'autre part au graphe $\mathcal{G}^{\mathfrak{M}^*}$:

$$\sum_{\kappa^* \sim \mathcal{L}^*} (u_{\kappa^*} - u_{\mathcal{L}^*})^2 \geq \frac{h_{\mathcal{G}^{\mathfrak{M}^*}}^2}{2} \sum_{\kappa^* \in \mathfrak{M}^*} (u_{\kappa^*} - c^{\mathfrak{M}^*})^2 \deg(x_{\kappa^*}), \quad (\text{II.3.4})$$

où on a utilisé l'expression Equation (II.2.3) pour les *spectral gaps*. La somme des inégalités II.3.3 et II.3.4 donne le résultat. $\square$

Lemma II.3.7. Soient $\kappa \in \mathfrak{M}$ et $\kappa^* \in \mathfrak{M}^*$. Alors

$$\deg(x_\kappa) \geq \frac{2 \mathfrak{m}(\mathbf{K})}{\text{reg}(\mathcal{T})^2 \text{size}(\mathcal{T})^2}, \quad \deg(x_{\kappa^*}) \geq \frac{2 \mathfrak{m}(\mathbf{K}^*)}{\text{reg}(\mathcal{T})^2 \text{size}(\mathcal{T})^2}.$$

■ **Proof.** On raisonne sur une maille primale plate $\mathbf{K}$. Posons $n_\kappa \stackrel{\text{d\'ef.}}{=} \deg(x_\kappa)$. Chaque arête de $\mathbf{K}$ correspond à une maille primale voisine donc n_κ est le nombre d'arêtes de $\mathbf{K}$. Puisque la maille est étoilée par rapport à x_κ , on peut la découper en n_κ triangles d'intérieurs disjoints :

$$\mathbf{K} = \bigcup_{i=1}^{n_\kappa} \Delta(x_\kappa, s_i, s_{i+1}),$$

où $(s_i)_{1 \leq i \leq n_K}$ sont les sommets de la maille, ordonnés cycliquement avec la convention $s_{n_K+1} = s_1$. Si on note θ_i l'angle $s_i x_K s_{i+1}$, alors

$$m(K) = \frac{1}{2} \sum_{i=1}^{n_K} \underbrace{\|s_i - x_K\|}_{\leq d_K} \underbrace{\|s_{i+1} - x_K\|}_{\leq d_K} |\sin \theta_i| \leq \frac{1}{2} n_K d_K^2.$$

On obtient alors

$$\deg(x_K) \geq \frac{2 m(K)}{d_K^2} \quad (\text{II.3.5})$$

et le même raisonnement s'applique aux mailles duales plates :

$$\deg(x_{K^*}) \geq \frac{2 m(K^*)}{d_{K^*}^2}.$$

Dans la régularité $\text{reg}(\mathcal{T})$, le terme

$$\max_{K \in \mathfrak{M}} \max_{D \in \mathfrak{D}_K} \frac{d_K}{d_D}$$

permet d'avoir l'estimation

$$d_K \leq \text{reg}(\mathcal{T}) d_D d_D \leq \text{reg}(\mathcal{T}) \text{size}(\mathcal{T}) \quad (\text{II.3.6})$$

pour tout diamant incident à K . La même propriété vaut pour les mailles duales. La combinaison de Equations (II.3.5) and (II.3.6) donne le résultat. $\square$

Remark II.3.8. Avec les poids $\mu_K = \frac{m(K)}{2}$ et $\mu_{K^*} = \frac{m(K^*)}{2}$, les minoration deviennent

$$\deg(x_K) \geq \frac{4}{\text{reg}(\mathcal{T})^2 \text{size}(\mathcal{T})^2} \mu_K, \quad \deg(x_{K^*}) \geq \frac{4}{\text{reg}(\mathcal{T})^2 \text{size}(\mathcal{T})^2} \mu_{K^*}.$$

Les Lemmas II.3.6 and II.3.7 vont nous permettre de montrer le résultat suivant.

Proposition II.3.9. Pour tout $u^T \in \tilde{\mathbb{R}}^T$,

$$\mathcal{E}(u^T) \geq \frac{2 h_G \tau^2}{\text{reg}(\mathcal{T})^2 \text{size}(\mathcal{T})^2} \|u^T\|_{2,T}^2.$$

L'ensemble $\tilde{\mathbb{R}}^T$ défini par Equation (I.4.11) est l'ensemble des fonctions scalaires DDFV à moyenne nulle.

■ **Proof.** Soit $u^T \in \tilde{\mathbb{R}}^T$. Par définition, la moyenne géométrique $\sum_{K \in \mathfrak{M}} \mu_K u_K$ est nulle. De plus, elle minimise la fonction $c \mapsto \sum_{K \in \mathfrak{M}} \mu_K (u_K - c)^2$. Par conséquent,

$$\sum_{K \in \mathfrak{M}} \mu_K (u_K - c^{\mathfrak{M}})^2 \geq \sum_{K \in \mathfrak{M}} \mu_K u_K^2,$$

et en utilisant les estimations du Lemma II.3.7 on trouve

$$\sum_{K \in \mathfrak{M}} (u_K - c^{\mathfrak{M}})^2 \deg(x_K) \geq \frac{4}{\text{reg}(\mathcal{T})^2 \text{size}(\mathcal{T})^2} \sum_{K \in \mathfrak{M}} \mu_K u_K^2. \quad (\text{II.3.7})$$

Le même raisonnement sur le maillage dual donne

$$\sum_{K^* \in \mathfrak{M}^*} (u_{K^*} - c^{\mathfrak{M}^*})^2 \deg(x_{K^*}) \geq \frac{4}{\text{reg}(\mathcal{T})^2 \text{size}(\mathcal{T})^2} \sum_{K^* \in \mathfrak{M}^*} \mu_{K^*} u_{K^*}^2. \quad (\text{II.3.8})$$

On somme les inégalités II.3.7 et II.3.8 puis le Lemma II.3.6 et la définition de la norme $\|\cdot\|_{2,T}$ donnent le résultat. $\square$

Intéressons-nous maintenant à la majoration de $\mathcal{E}$.

Proposition II.3.10. Pour tout $u^T \in \mathbb{R}^T$,

$$\mathcal{E}(u^T) \leq 4 \operatorname{reg}(\mathcal{T})^2 \|\nabla^{\mathcal{D}} u^T\|_{2,\mathcal{D}}^2.$$

■ **Proof.** Soient deux mailles primales voisines $\kappa \sim \mathcal{L}$ définissant le diamant $\mathcal{D} = \mathcal{D}_{\sigma',\sigma^*}$. Par définition du gradient I.4.1 sur $\mathcal{D}$, on a

$$(u_{\mathcal{L}} - u_{\kappa})^2 = m(\sigma^*)^2 |(\nabla^{\mathcal{D}} u^T, \tau_{\kappa,\mathcal{L}})|^2 \leq m(\sigma^*)^2 |\nabla^{\mathcal{D}} u^T|^2.$$

De même, pour deux mailles duales voisines $\kappa^* \sim \mathcal{L}^*$ définissant le diamant $\mathcal{D} = \mathcal{D}_{\sigma,\sigma^{*'}}$, on a

$$(u_{\mathcal{L}^*} - u_{\kappa^*})^2 = m(\sigma)^2 |(\nabla^{\mathcal{D}} u^T, \tau_{\kappa^*,\mathcal{L}^*})|^2 \leq m(\sigma)^2 |\nabla^{\mathcal{D}} u^T|^2.$$

Les ensembles $\mathfrak{E}$ et $\mathfrak{E}^*$ sont en bijection avec l'ensemble $\mathfrak{D}$. Ainsi

$$\mathcal{E}(u^T) \leq \sum_{\mathcal{D} \in \mathfrak{D}} [m(\sigma)^2 + m(\sigma^*)^2] |\nabla^{\mathcal{D}} u^T|^2. \quad (\text{II.3.9})$$

On souhaite avoir $m(\sigma)^2 + m(\sigma^*)^2 \leq C \mu_{\mathcal{D}}$ avec $C > 0$ indépendante de $\operatorname{size}(\mathcal{T})$. Comme

$$\frac{m(\sigma)^2 + m(\sigma^*)^2}{\mu_{\mathcal{D}}} = \frac{2}{\sin \theta_{\mathcal{D}}} \left(\frac{m(\sigma)}{m(\sigma^*)} + \frac{m(\sigma^*)}{m(\sigma)} \right), \quad (\text{II.3.10})$$

on met dans la régularité

$$\max_{\mathcal{D}_{\sigma,\sigma^*} \in \mathfrak{D}} \left\{ \frac{m(\sigma)}{m(\sigma^*)}, \frac{m(\sigma^*)}{m(\sigma)} \right\} \quad \text{et} \quad \frac{1}{\sin \theta_{\mathcal{T}}}.$$

Autrement dit, il faut imposer que l'angle entre deux diagonales ne tende ni vers 0 ni vers π et que le rapport des longueurs des deux diagonales reste uniformément borné.

Remark II.3.11. Ceci est contrôlé par

$$\max_{\mathcal{D} \in \mathfrak{D}} \max_{\mathcal{Q} \in \Omega_{\mathcal{D}}} \frac{d_{\mathcal{D}}}{\min_{\eta \in \partial \mathcal{Q}} m(\eta)}.$$

Ainsi, d'après Equation (II.3.10),

$$m(\sigma)^2 + m(\sigma^*)^2 \leq 4 \operatorname{reg}(\mathcal{T})^2 \mu_{\mathcal{D}}$$

ce qui, avec II.3.9, fournit le résultat. □

Les Propositions II.3.9 and II.3.10 aboutissent à l'inégalité

$$\forall u^T \in \tilde{\mathbb{R}}^T, \quad \frac{1}{\sqrt{2} \operatorname{reg}(\mathcal{T})^2} \frac{h_{\mathcal{G}^T}}{\operatorname{size}(\mathcal{T})} \|u^T\|_{2,\mathcal{T}} \leq \|\nabla^{\mathcal{D}} u^T\|_{2,\mathcal{D}}.$$

Pour l'instant, la régularité $\operatorname{reg}(\mathcal{T})$ contrôle la géométrie locale des mailles. Elle autorise cependant les raffinements locaux. Elle ne contrôle donc pas le rapport entre la plus grande et la plus petite taille de maille.

Dans la suite, nous allons démontrer que sous des hypothèses géométriques appropriées, il existe une constante $c > 0$ indépendante de $\operatorname{size}(\mathcal{T})$ telle que

$$h_{\mathcal{G}^T} \geq c \operatorname{size}(\mathcal{T}).$$

Pour cela, nous allons ajouter une condition de quasi-uniformité globale du type

$$\vartheta \operatorname{size}(\mathcal{T}) \leq d_{\mathcal{D}} \quad \forall \mathcal{D} \in \mathfrak{D}.$$

Proposition II.3.12. On suppose qu'il existe $N, c_A, C_{\sigma} > 0$ indépendantes de $\operatorname{size}(\mathcal{T})$ telles que

$$\deg(x_{\kappa}) \leq N, \quad m(\kappa) \geq c_A \operatorname{size}(\mathcal{T})^2, \quad m(\sigma) \leq C_{\sigma} \operatorname{size}(\mathcal{T}).$$

Alors, pour tout $W \subset V(\mathcal{G}^{\mathfrak{M}})$, on a

$$\min\{\operatorname{vol}(W), \operatorname{vol}(W^c)\} \leq \frac{N}{c_A \operatorname{size}(\mathcal{T})^2} \min\{m(A_W), m(\mathcal{S} \setminus A_W)\},$$

où

$$A_W = \bigcup_{x_\kappa \in W} \kappa \subset \mathcal{S}.$$

■ **Proof.** Puisque les mailles forment une partition,

$$m(A_W) = \sum_{x_\kappa \in W} m(\kappa).$$

Le volume combinatoire vaut

$$\text{vol}(W) = \sum_{x_\kappa \in W} \deg(x_\kappa).$$

Or, pour chaque maille,

$$\deg(x_\kappa) \leq N \leq \frac{N}{c_A \text{size}(\mathcal{T})^2} m(\kappa).$$

En sommant,

$$\text{vol}(W) \leq \frac{N}{c_A \text{size}(\mathcal{T})^2} m(A_W).$$

De même,

$$\text{vol}(W^c) \leq \frac{N}{c_A \text{size}(\mathcal{T})^2} m(\mathcal{S} \setminus A_W).$$

On en déduit

$$\min\{\text{vol}(W), \text{vol}(W^c)\} \leq \frac{N}{c_A \text{size}(\mathcal{T})^2} \min\{m(A_W), m(\mathcal{S} \setminus A_W)\}.$$

Pour obtenir $m(\kappa) \geq c_A \text{size}(\mathcal{T})^2$, on peut combiner

$$\frac{d_\kappa}{\sqrt{m(\kappa)}} \leq \text{reg}(\mathcal{T})$$

avec une quasi-uniformité globale

$$d_\kappa \geq \vartheta \text{size}(\mathcal{T}).$$

□

Lemma II.3.13.

$$|\partial W| \geq \frac{c}{\text{size}(\mathcal{T})} m(\partial A_W).$$

Stratégie. Il faut alors établir deux comparaisons uniformes:

$$\min\{\text{vol}(W), \text{vol}(W^c)\} \leq \frac{C}{\text{size}(\mathcal{T})^2} \min\{m(A_W), m(\mathcal{S} \setminus A_W)\},$$

et

$$|\partial W| \geq \frac{c}{\text{size}(\mathcal{T})} m(\partial A_W).$$

Elles donnent

$$\frac{|\partial W|}{\min\{\text{vol}(W), \text{vol}(W^c)\}} \geq c \text{size}(\mathcal{T}) \frac{m(\partial A_W)}{\min\{m(A_W), m(\mathcal{S} \setminus A_W)\}} \geq c \text{size}(\mathcal{T}) h(\mathcal{S}).$$

En prenant l'inf sur W , on obtient

$$h_{\mathcal{G}^{\mathfrak{M}}} \geq c \text{size}(\mathcal{T}) h(\mathcal{S}).$$

Comme $\mathcal{S}$ est une surface compacte connexe fixée, $h(\mathcal{S}) > 0$, donc

$$h_{\mathcal{G}^{\mathfrak{M}}} \geq c_0 \text{size}(\mathcal{T}).$$

Proposition II.3.14.

$$\|\Pi_T^{\mathcal{M}} u^T\|_{L^2(\mathcal{M})}^2 \sim h^2 \sum_{\kappa \in \mathfrak{M}} u_\kappa^2$$

II.3.3 Passage du discret au continu

Definition II.3.15. Constante isopérimétrique

Lemma II.3.16. L'inégalité ?? est vraie pour $h = h(\mathcal{G})$.

Lemma II.3.17. Soit $A \subset \mathcal{M}$,

$$\inf \frac{|\partial A|}{|A|} \dots$$

Lemma II.3.18.

$$\frac{|\partial S|}{|S|} \sim \frac{|\partial A|}{|A|} h$$

Proposition II.3.19 (Relaxation condition de Cheeger). Pour $0 < k < 1$, on définit

$$h_k(\mathcal{M}) \stackrel{\text{déf.}}{=} \inf_{\substack{A \subset \mathcal{M} \\ |A| \leq k |\mathcal{M}|}} \frac{|\partial A|}{|A|},$$

de sorte que $h(\mathcal{M}) = h_{1/2}(\mathcal{M})$. Alors

$$\begin{cases} h_k(\mathcal{M}) = h(\mathcal{M}) & \text{si } \frac{1}{2} \leq k < 1 \text{ (relaxation),} \\ h_k(\mathcal{M}) \geq h(\mathcal{M}) & \text{si } 0 < k < \frac{1}{2} \text{ (restriction).} \end{cases}$$

■ **Proof.** On remarque que pour $0 < k \leq k' < 1$, alors $h_{k'}(\mathcal{M}) \leq h_k(\mathcal{M})$ car on minimise sur un ensemble plus grand.
 $h(\mathcal{M}) = h_{1/2}(\mathcal{M})$ □

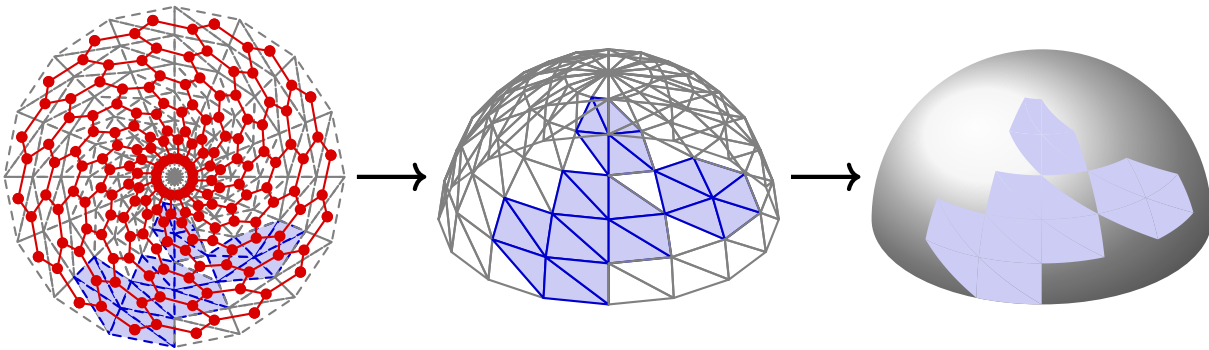


Figure II.3. maillage non régulier

Sketch of the proof.

- La courbure de la surface est minorée et donc dès que h est suffisamment petit, tous les triangles sont dans un voisinage tubulaire de la surface qui est tel qu'on peut toujours projeter de manière unique dessus (de tout point du maillage, on peut projeter). On peut projeter les maillages sur la surface et on peut le faire de sorte que les volumes restent en h^2 des triangles et les longueurs des arêtes restent en h et peut être que les angles soient minorés par une constante uniforme indépendamment de h (on peut l'avoir avec la contrainte sur la régularité du maillage).

- Les diamants doivent aussi être proches de la surface (il faudra projeter les diamants et estimer)
- Le u est de deux sortes: P^0 du maillage duale.
- On veut une inégalité comme:

$$\left\| \tilde{\nabla} u \right\|_{L^2} \geq C \|u\|_{L^2}$$

- Tout le but de l'opération est de montrer que

$$\boxed{\sum_{i \sim j} (u_i - u_j)^2 \geq C h^2 \sum (u_i - c)^2} \quad (\text{II.3.11})$$

Il faut minimiser sur c , prendre la meilleure constante, la projection de u sur les constantes.

- **II.3.11** est une inégalité de graphe, c'est le laplacien sur les graphes qui relie le membre de gauche au membre de droite avec une constante qui est la plus petite valeur propre du laplacien sur le graphe. **Il faut estimer cette constante.**
- Ceci est vrai avec $h = h(G)$ (Cheeger) avec

$$h(G) \stackrel{\text{d\'ef.}}{=} \inf_{|S| \leq \frac{|T_h|}{2}} \frac{|\partial S|}{|S|}$$

- Il y a autant d'arêtes qui relient S aux autres que d'arêtes de bord qui contiennent les points de S .
- Tout le boulot est de montrer que

$$h(G) \geq h c$$

- Soit S . On considère l'ensemble de tous les triangles qui sont de centre un point de S et on projette ces triangles sur la surface.
- On considère le sous-ensemble de la surface qui est le projeté des triangles du maillage pour lesquels il y a un point de S qui est dedans. On appelle $A \subset \mathcal{M}$ cet ensemble.

$$A = \bigcup_{i \in S} \Delta_i^{\mathcal{M}}.$$

- D'après les lemmes de projection et les hypothèses : $|\partial S| \sim \frac{|\partial A|}{h}$ et $|S| \sim \frac{|A|}{h^2}$ et donc

$$\frac{|\partial S|}{|S|} \sim \frac{|\partial A|}{|A|} h$$

- $\frac{|\partial A|}{|A|}$: on est sur la surface, c'est la constante d'isopérimétrie de la variété et c'est minoré par quelque chose de strictement positif avec la condition $\boxed{|A| \leq \frac{|\mathcal{M}|}{2}}$.

II.3.4 État de l'art

Part II

Appendices

Appendix A

Simplicial Complexes

<https://team.inria.fr/geometrica/files/2015/11/1-simplicial-complexes.pdf>

A.1 Geometric simplices

Definition A.1.1. A k -simplex σ is the convex hull of $k + 1$ points of $\mathbb{R}^d$ that are affinely independent.

$$\sigma = \text{conv}(p_0, \dots, p_k) = \left\{ x \in \mathbb{R}^d, x = \sum_{i=0}^k \lambda_i p_i, \lambda_i \in [0; 1], \sum_{i=0}^k \lambda_i = 1 \right\}$$

$k = \dim(\text{aff}(\sigma))$ is called the dimension of σ .

1-simplex = line segment, 2-simplex = triangle, 3-simplex = tetrahedron

$V(\sigma)$ = set of vertices of a k -simplex σ

$\forall V' \subset V(\sigma)$, $\text{conv}(V')$ is a face of σ .

A.1.1 Geometric simplicial complexes

A finite collection of simplices K called the faces of K such that

- $\forall \sigma \in K$, σ is a simplex
 - $\sigma \in K, \tau \subset \sigma \implies \tau \in K$
 - $\forall \sigma, \tau \in K$, either $\sigma \cap \tau = \emptyset$ or $\sigma \cap \tau$ is a common face of both
- The dimension of a simplicial complex K is the max dimension of its simplices.
The underlying space $|K| \subset \mathbb{R}^d$ of K is the union of the simplices of K .

A.1.2 Triangulation of a finite point set of $\mathbb{R}^d$

- A simplicial d -complex K is pure if every simplex in K is the face of a d -simplex.
- A triangulation of a finite point set $P \in \mathbb{R}^d$ is a pure geometric simplicial complex K s.t. $\text{vert}(K) = P$ and $|K| = \text{conv}(P)$

A.1.3 Triangulation of a polygonal domain of $\mathbb{R}^2$

A triangulation of a polygonal domain $\Omega \in \mathbb{R}^2$ is a pure geometric simplicial complex K s.t. $\text{vert}(K) = \text{vert}(\Omega)$ and $|K| = \Omega$

Appendix B

Notations

Lisse			Polyédrique		
$\mathcal{S}$	Surface fermée de $\mathbb{R}^3$	— $\Pi^{\mathcal{S}} : \mathcal{S} \rightarrow \mathcal{S}$	S	Approximation polyédrique de $\mathcal{S}$ de pas h	—
Primal					
$\mathfrak{M}$	Maillage primal	—	$\mathbf{M}$	Maillage primal de S	—
$\kappa, \mathcal{L}$	Mailles primales	—	K, L	Mailles primales plates	—
$m(\kappa)$	Mesure de la maille κ	—	$m(K)$	Mesure de la maille plate K	—
d_{κ}	Diamètre de κ	—	d_K	Diamètre de K	—
ρ_{κ}	Rayon de κ	—	ρ_K	Rayon de K	—
	$\mathfrak{B}^* = \{x_{\kappa^*} \mid \kappa^* \in \mathfrak{M}^*\}$		Ensemble des sommets de $\mathfrak{M}$ et de $\mathbf{M}$		
ς		—	σ	Arête primale	—
$m(\sigma)$	Longueur de l'arête σ	—			—
$\mathfrak{E}$	Ensemble des arêtes primales	—	$\mathbf{E}$		—
$\mathfrak{E}_{\kappa}$	Ensemble des arêtes de κ	—			—
			$h_{\mathbf{M}}$		I.1.2
Dual					
$\mathfrak{M}^*$	Maillage dual	—			—
$\kappa^*, \mathcal{L}^*$	Mailles duales	—	K^*, L^*	Mailles duales plates	—
$m(\kappa^*)$	Mesure de la maille κ^*	—			—
d_{κ^*}	Diamètre de κ^*	—			—
$\mathfrak{B}$	Ensemble des sommets de $\mathfrak{M}^*$	—	$\{x_{\kappa} \mid \kappa \in \mathfrak{M}\}$		—
$\sigma^* = [x_{\kappa}, x_{\mathcal{L}}]$	Arête duale	—			—
$m(\sigma^*)$	Longueur de l'arête σ^*	—			—
$\mathfrak{E}^*$	Ensemble des arêtes duales	—			—
$\mathfrak{E}_{\kappa^*}^*$	Ensemble des arêtes de κ^*	—			—
Diamant					
$\mathcal{D} = \mathcal{D}_{\sigma, \sigma^*}$	Diamant	I.2.3	D		—
$\mathfrak{D}$	Maillage diamant	I.2.3			—
$\mathfrak{D}_{\kappa}$	$\{\mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D} \mid \sigma \in \mathfrak{E}_{\kappa}\}$	I.2.3			—
$\mathfrak{D}_{\kappa^*}$	$\{\mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D} \mid \sigma^* \in \mathfrak{E}_{\kappa^*}^*\}$	I.2.3			—
$\nu_{\sigma \kappa}$	Normale unitaire à σ sortante de κ	I.8a			—
$\tau_{\kappa^*, \mathcal{L}^*}$	Tangente unitaire à σ orientée de κ^* à $\mathcal{L}^*$	I.8a			—
$\nu_{\sigma^* \kappa^*}$	Normale unitaire à σ^* sortante de κ^*	I.8a			—
$\tau_{\kappa, \mathcal{L}}$	Tangente unitaire à σ^* orientée de κ à $\mathcal{L}$	I.8a			—
$\theta_{\mathcal{D}}$	Angle entre σ et σ^*	I.8a			—

Table B.1. Notations générales des maillages DDFV

Titre	
$\mathcal{T} = (\mathfrak{M}, \mathfrak{M}^*)$	Maillage DDFV
$\text{size}(\mathcal{T})$	
$\text{reg}(\mathcal{T})$	Régularité géométrique d'un maillage $\mathcal{T}$
u_{κ}	Inconnue sur la maille primale κ
u_{κ^*}	Inconnue sur la maille duale κ^*
$u^{\mathfrak{M}}$	$(u_{\kappa})_{\kappa \in \mathfrak{M}}$
$u^{\mathfrak{M}^*}$	$(u_{\kappa^*})_{\kappa^* \in \mathfrak{M}^*}$
$u^{\mathcal{T}}$	Inconnues u_{κ} pour tout $\kappa \in \mathfrak{M}$ et u_{κ^*} pour tout $\kappa^* \in \mathfrak{M}^*$ I.3.1
	$u^{\mathcal{T}} = (u^{\mathfrak{M}}, u^{\mathfrak{M}^*})$
$\mathbb{R}^{\mathfrak{M}}$	Ensemble des inconnues $u^{\mathfrak{M}}$
$\mathbb{R}^{\mathfrak{M}^*}$	Ensemble des inconnues $u^{\mathfrak{M}^*}$
$\mathbb{R}^{\mathcal{T}}$	Ensemble des inconnues $u^{\mathcal{T}}$
$(\mathbb{R}^2)^{\mathfrak{D}}$	Ensemble des vecteurs de $\mathbb{R}^2$ définis par diamant $\mathfrak{D}$
$\xi^{\mathfrak{D}}, \phi^{\mathfrak{D}}$	Inconnues de $(\mathbb{R}^2)^{\mathfrak{D}}$
$(u^{\mathcal{T}}, v^{\mathcal{T}})_{\mathcal{T}}$	I.4.7
$(\xi^{\mathfrak{D}}, \phi^{\mathfrak{D}})_{\mathfrak{D}}$	I.4.8
$\mu_{\kappa}, \mu_{\kappa^*}$	Poids pour $(\cdot, \cdot)_{\mathcal{T}}$
$\mu_{\mathfrak{D}}$	Poids pour $(\cdot, \cdot)_{\mathfrak{D}}$
$\widetilde{\mathbb{R}}^{\mathcal{T}}$	
$\ u^{\mathcal{T}}\ _{2,\mathcal{T}}$	$(u^{\mathcal{T}}, u^{\mathcal{T}})_{\mathcal{T}}^{1/2}$ pour tout $u^{\mathcal{T}} \in \mathbb{R}^{\mathcal{T}}$
$\ u^{\mathcal{T}}\ _{p,\mathcal{T}}$	I.4.9
$\ \xi^{\mathfrak{D}}\ _{2,\mathfrak{D}}$	$(\xi^{\mathfrak{D}}, \xi^{\mathfrak{D}})_{\mathfrak{D}}^{1/2}$ pour tout $\xi^{\mathfrak{D}} \in (\mathbb{R}^2)^{\mathfrak{D}}$
$\Pi_{\mathcal{T}}^{\mathcal{S}}$	$\Pi_{\mathcal{T}}^{\mathcal{S}}: \mathbb{R}^{\mathcal{T}} \rightarrow \text{L}^2(\mathcal{S})$ I.3.2
$\Pi_{\mathfrak{D}}^{\mathcal{S}}$	$\Pi_{\mathfrak{D}}^{\mathcal{S}}: (\mathbb{R}^2)^{\mathfrak{D}} \rightarrow \text{L}^2(\mathcal{S})$ I.3.3

Table B.2. Notations

Opérateurs	
$\nabla^{\mathfrak{D}}$	Opérateur gradient discret de $\mathbb{R}^{\mathcal{T}} \rightarrow (\mathbb{R}^2)^{\mathfrak{D}}$ I.4.1
$\nabla^{\mathfrak{D}} u^{\mathcal{T}}$	Gradient discret de $u^{\mathcal{T}}$ sur le diamant $\mathfrak{D}$ I.4.2
$\mathbf{rot}^{\mathfrak{D}}$	Opérateur rotationnel vectoriel discret de $\mathbb{R}^{\mathcal{T}} \rightarrow (\mathbb{R}^2)^{\mathfrak{D}}$ I.4.5
$\text{div}^{\mathcal{T}}$	Opérateur divergence discrète de $(\mathbb{R}^2)^{\mathfrak{D}} \rightarrow \mathbb{R}^{\mathcal{T}}$
$\text{rot}^{\mathcal{T}}$	Opérateur rotationnel scalaire discret de $(\mathbb{R}^2)^{\mathfrak{D}} \rightarrow \mathbb{R}^{\mathcal{T}}$

Matrices

$\mathbf{G}_{\mathfrak{D}}$	Matrice du gradient
$\mathbf{D}_{\mathcal{T}}$	Matrice de la divergence
$\mathbf{R}_{\mathfrak{D}}$	
$M_{\mathfrak{D}}$	Poids
$M_{\mathcal{T}}$	Poids

Graphes

$\mathcal{G}^{\mathfrak{M}}$	Graphe associé au maillage $\mathfrak{M}$	$(\mathfrak{B}, \mathfrak{E}^*)$
$\mathcal{G}^{\mathfrak{M}^*}$	Graphe associé au maillage $\mathfrak{M}^*$	$(\mathfrak{B}^*, \mathfrak{E})$
$\deg v$	Degré du sommet v	—
$\text{vol } S$	Volume d'un ensemble de sommets S	—

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